BIOMASS applications, challenges and new opportunities



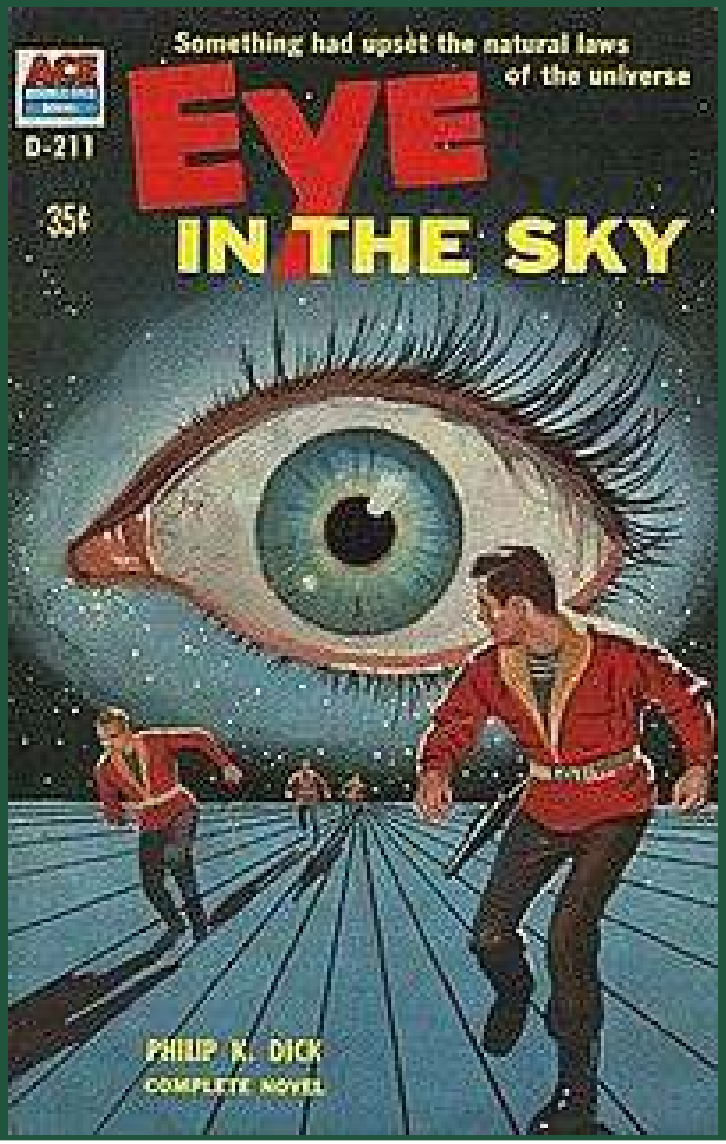
Armando Marino
The University of Stirling, Scotland

Subtitle:

The P-band eye in the sky

What is this not about… unfortunately

I am afraid we are NOT going to talk about sci-fi literature…



The Eye in the Sky, Philip Dick (1957)

What is this not about…

… also we will NOT talk about spying and Cold War…



Dr. Strangelove or: How I Learned to Stop Worrying and Love the Bomb, Stanley Kubrick (1964)

What is this not about… gladly

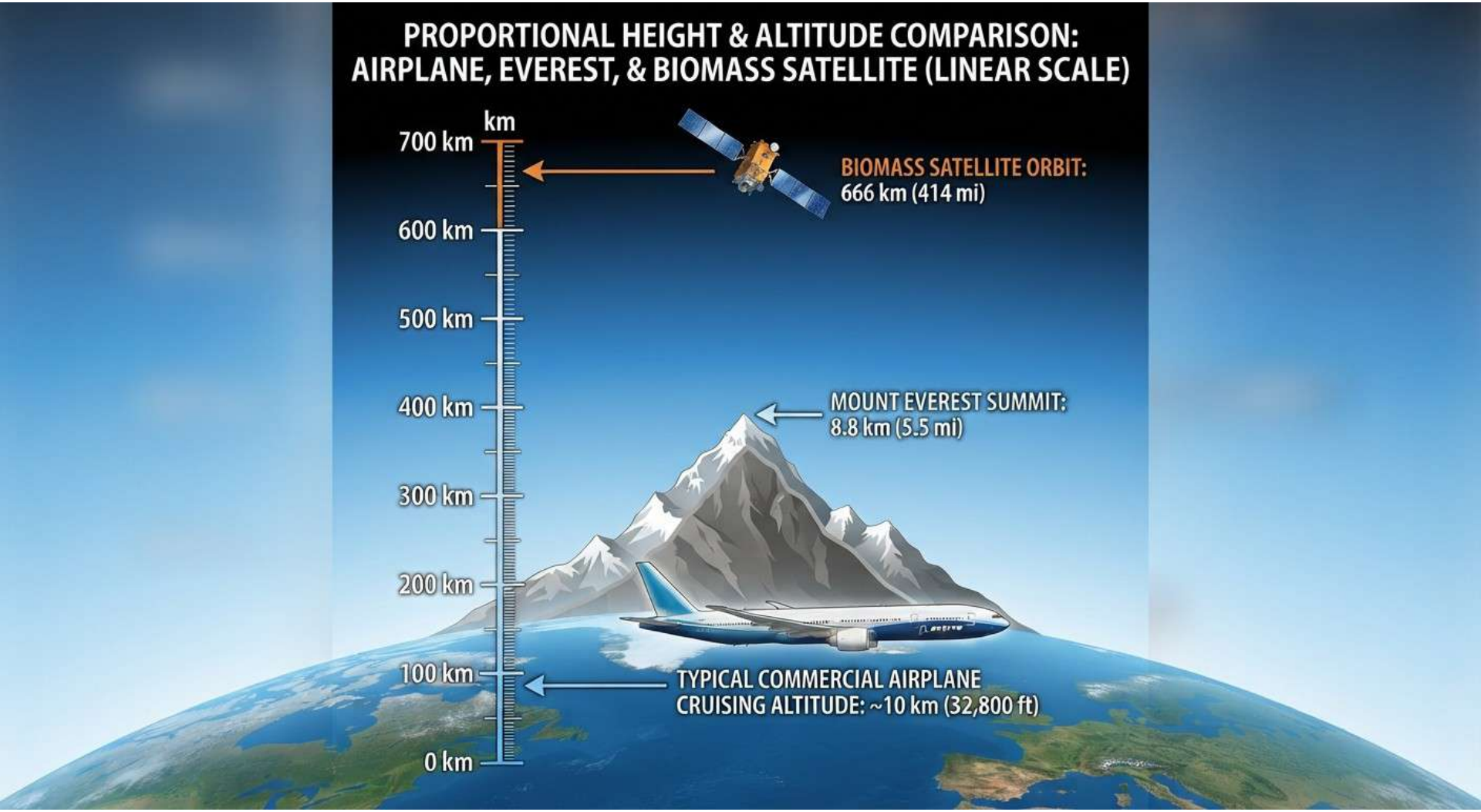
Skitters, Falling skies (TV Series)



…and I am very glad we will NOT talk about alien invasions…

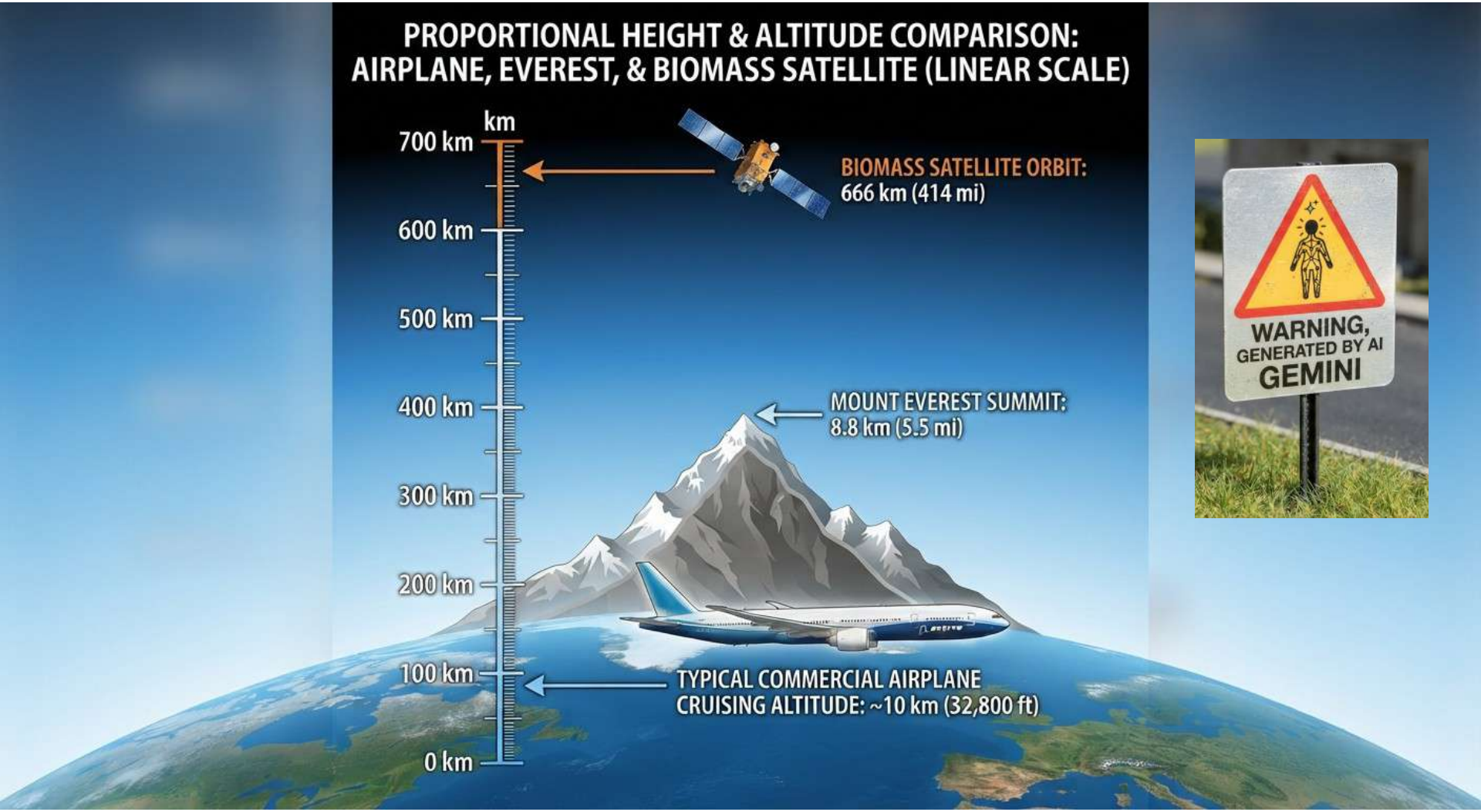
Earth Observation satellites fly above us

Space… next frontier…

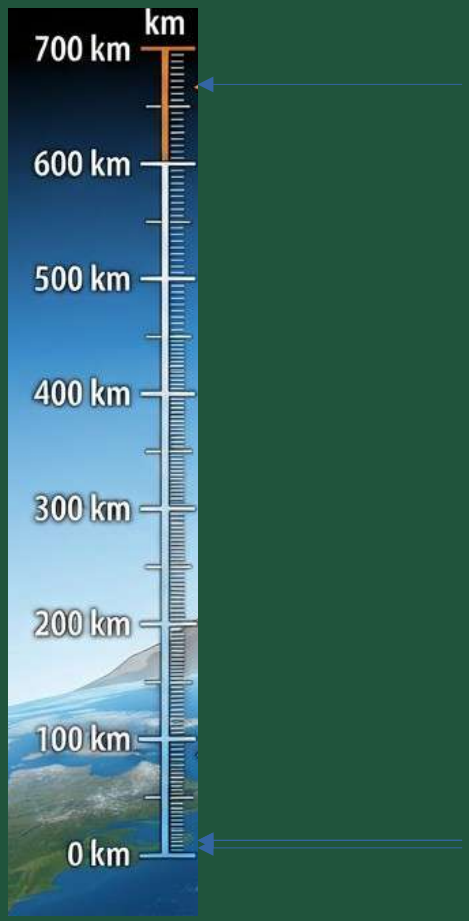


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Space… next frontier…

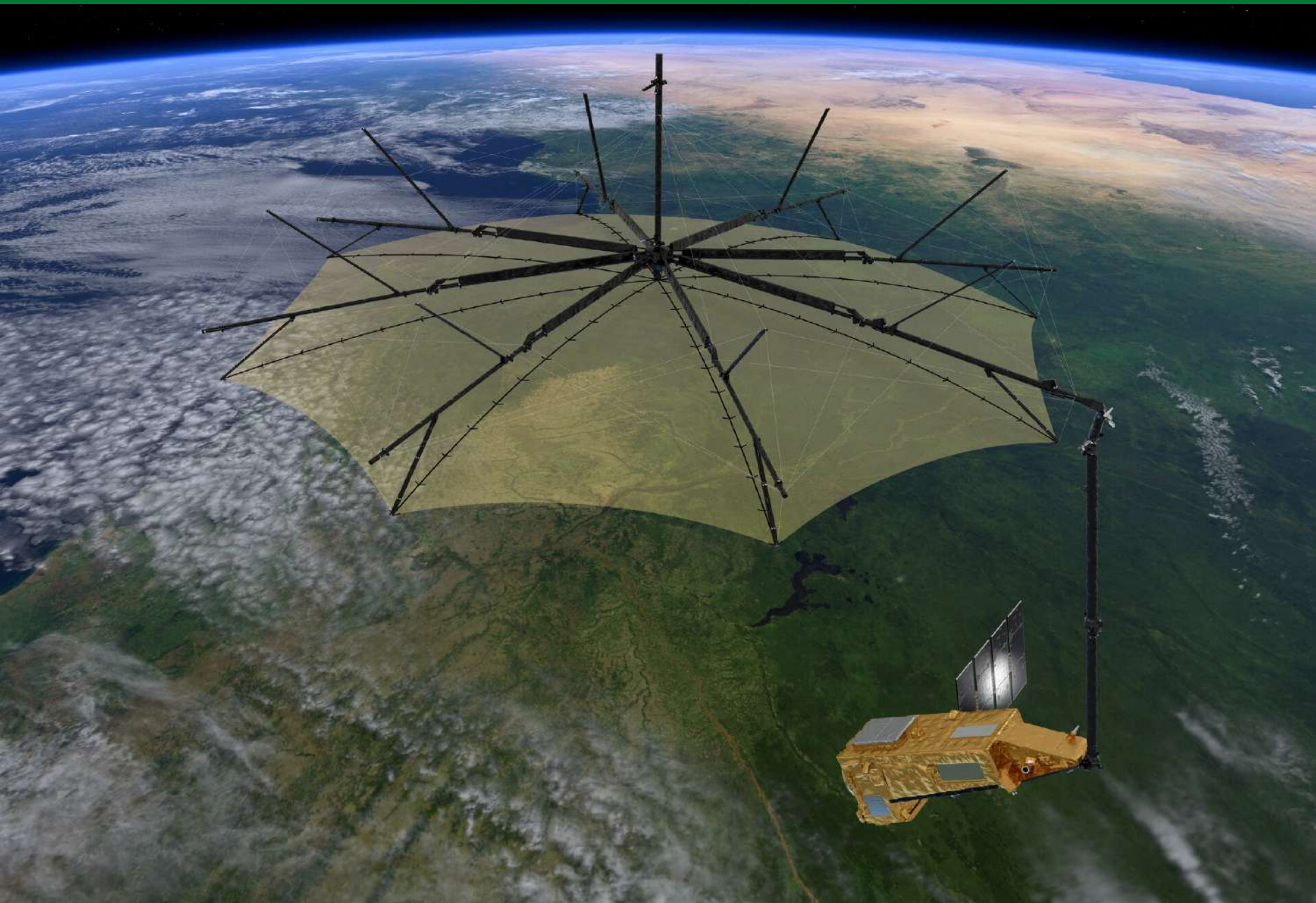


Space… with correct proportions

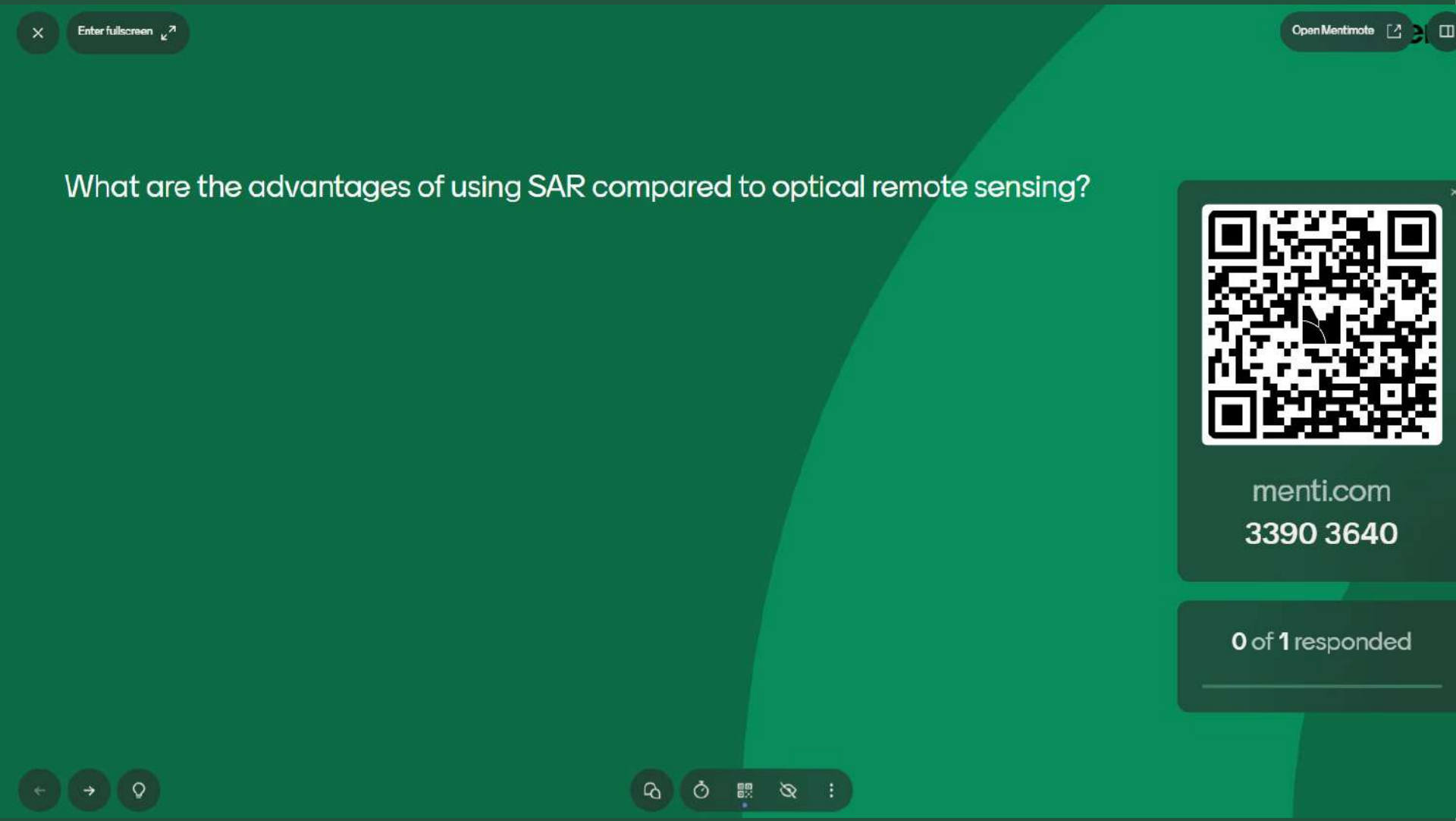


- ✓ This are the correct proportions
- ✓ It is very high…

A P-band “eye” flying above our heads



Initial survey



SAR magic

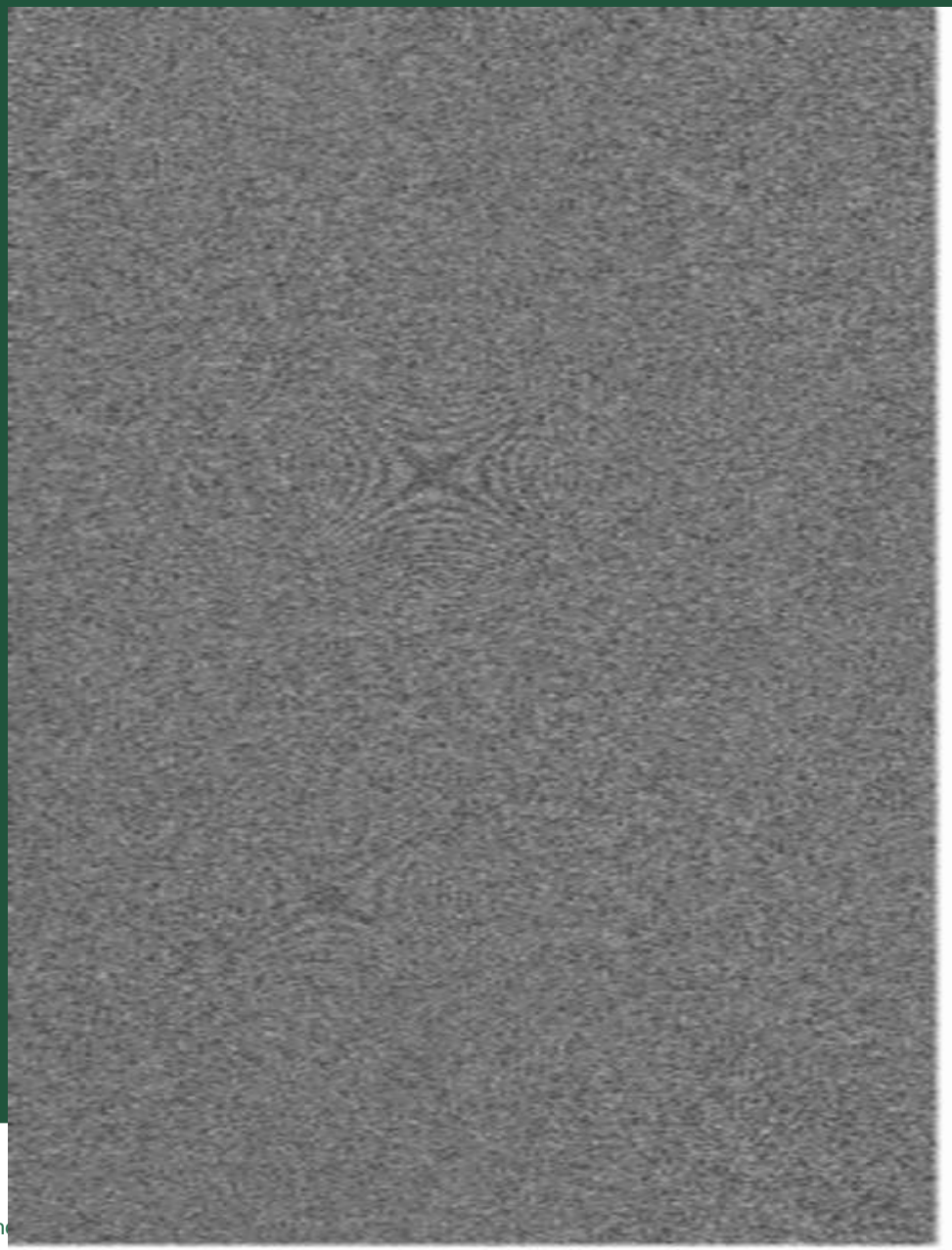
SAR magic



SAR magic: row data

- ✓ The row data are voltages collected by antennas in space
- ✓ It looks like noise when we acquire it.
- ✓ Using focusing, from noise…

https://www.esa.int/Enabling_Support/Space_Engineering_Technology/Onboard_Data_Processing/Introduction_to_a_SAR_System



SAR magic

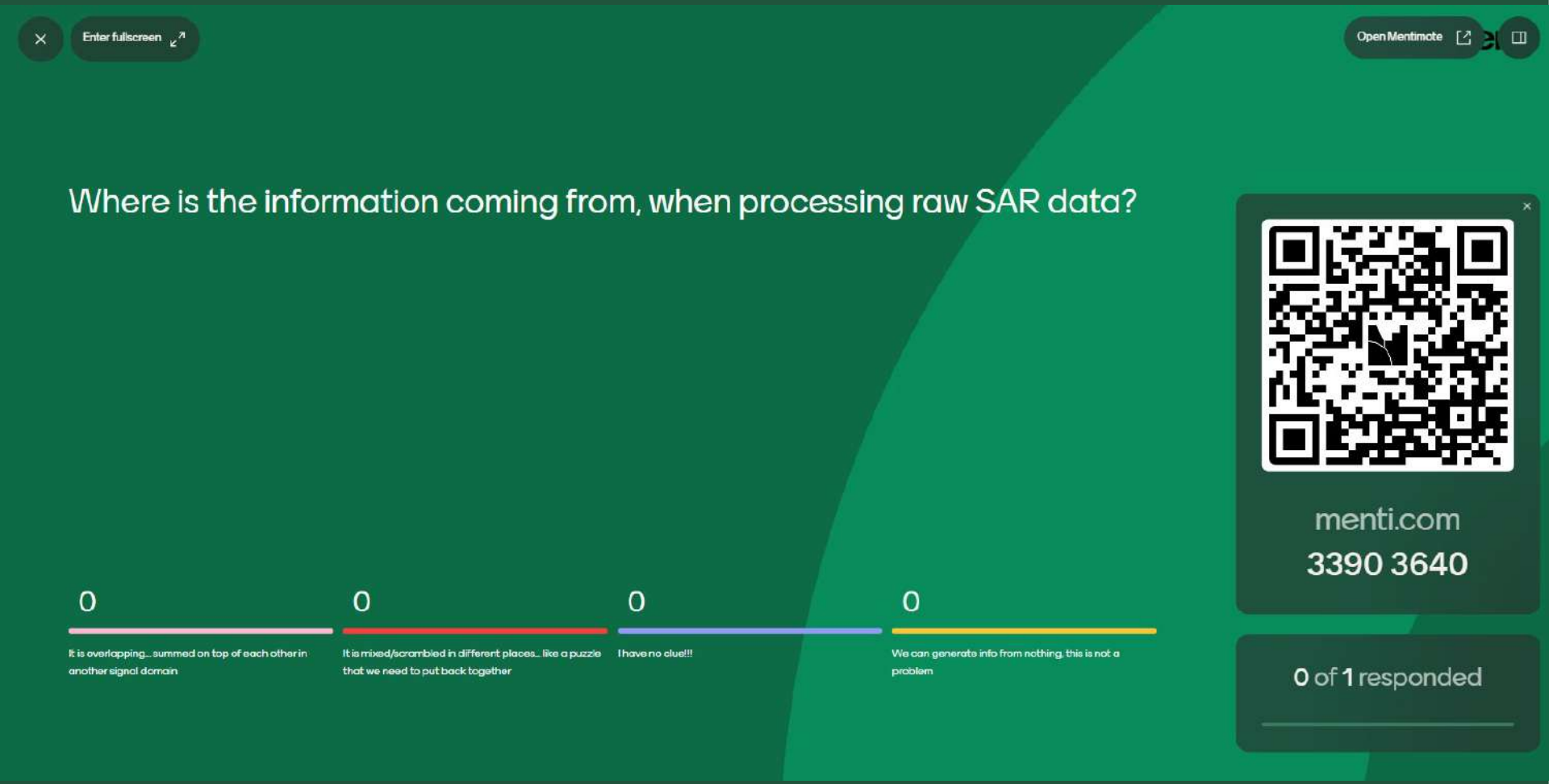
- ✓ From noise we make an image.
- ✓ Where is this information coming from? How can we make something from what seems nothing?

https://www.esa.int/Enabling_Support/Space_Engineering_Technology/Onboard_Data_Processing/Introduction_to_a_SAR_System



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Generating info from nothing?!?!?



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Moral of the story...

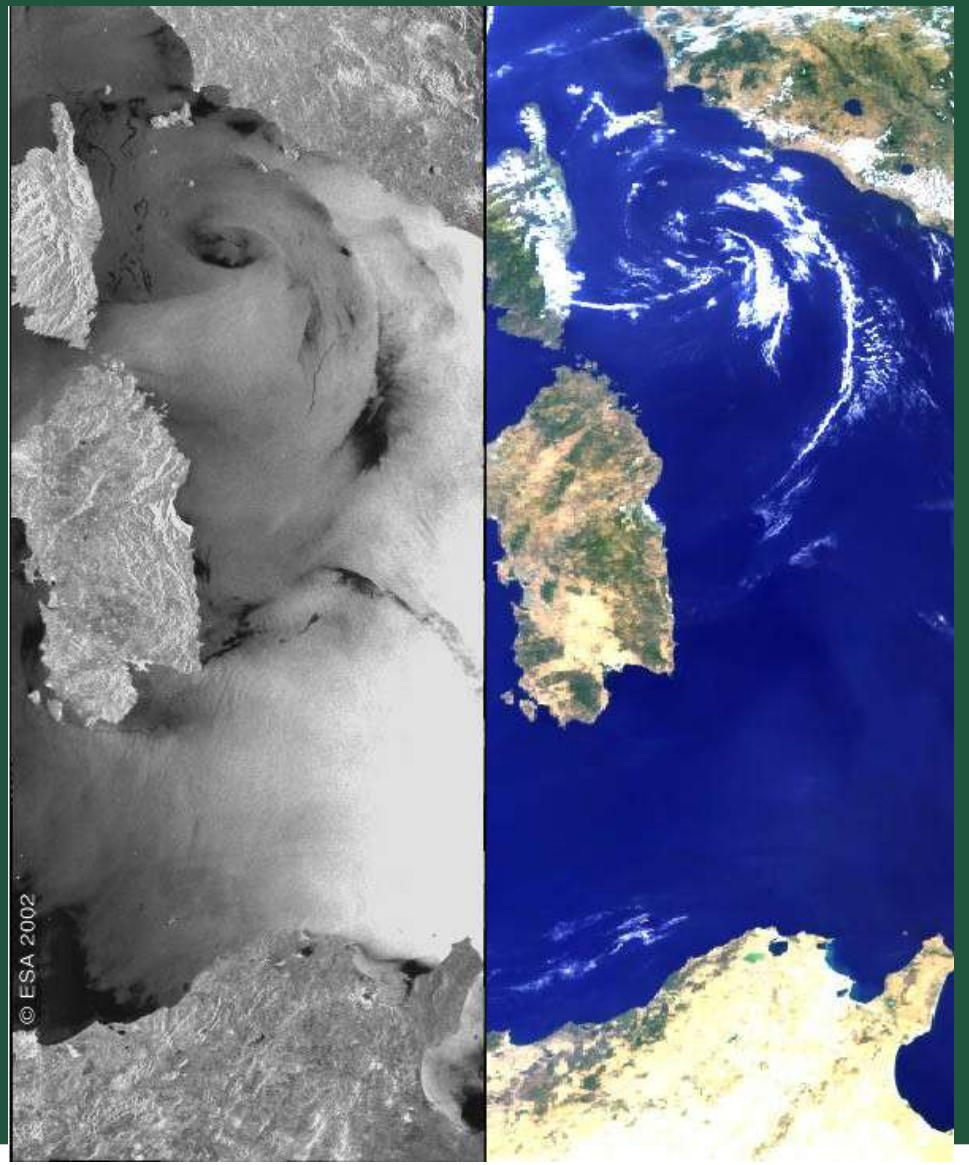
- ✓ Sometimes you may not see things clearly in a SAR image… but this does not mean the information is not there...
- ✓ Our brain is limited when it comes to analysis multi-dimensional data, especially if these are in complex number format.
- ✓ Maths and algorithms can help us see things though…

DO NOT GIVE UP IF YOU DO NOT SEE WHAT YOU LIKE IN THE IMAGE

SAR properties: clouds

Going though clouds

- ✓ “Seeing” through clouds is the first thing that comes to mind when talking about SAR
- ✓ But I like this image for another characteristic… this shows another great advantage of using SAR compared to optical...



Why I like the previous example?

What other great property of SAR is shown there?



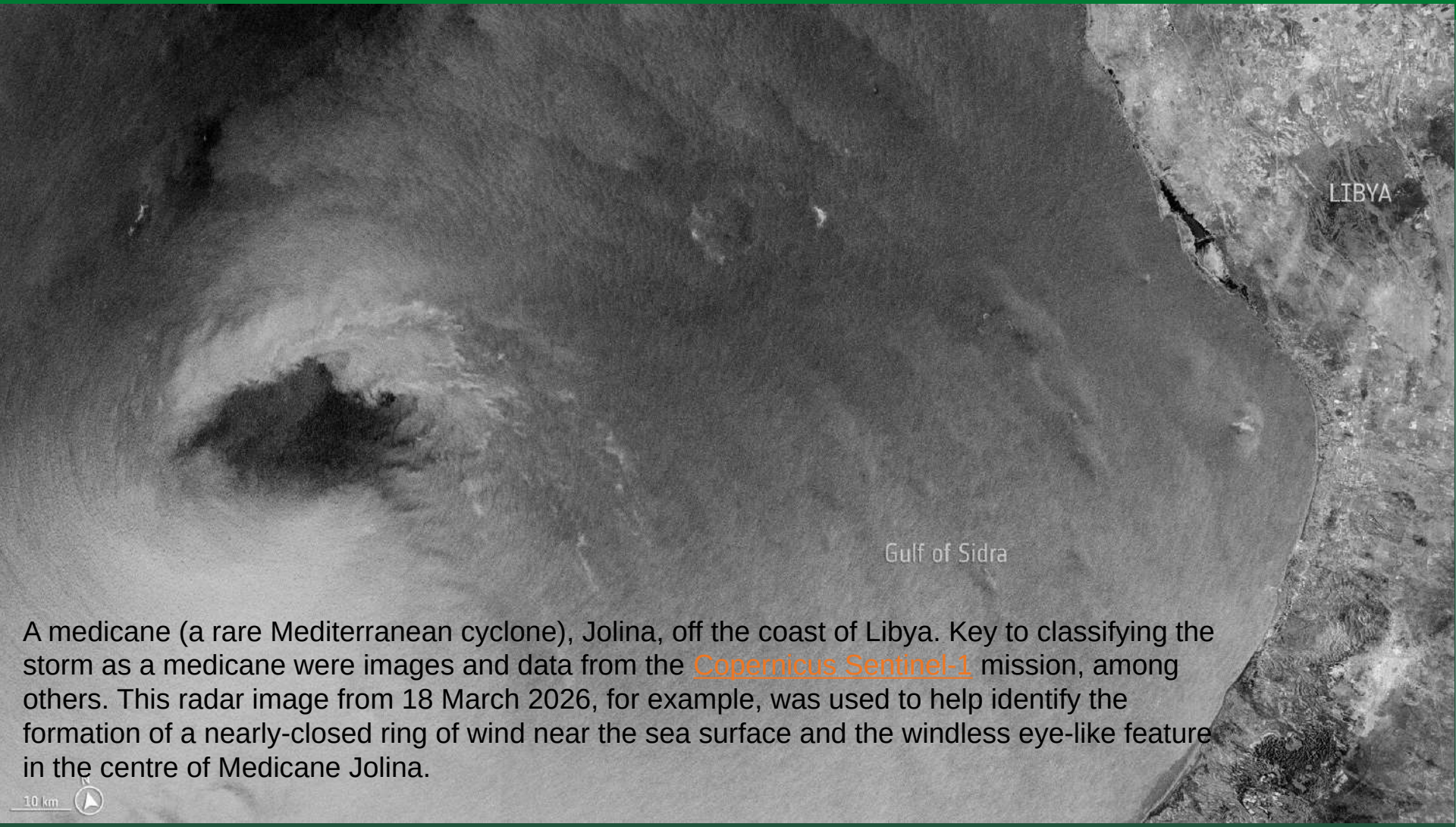
Moral of the story...

- ✓ SAR is really good to provide timely images, independent on clouds.
- ✓ It is very good to monitor cloudy places, like BIOMASS is doing.

EXPLOIT THIS CAPABILITY BY THINK OF SAR WHEN IT COMES TO CLOUDY PLACES/APPLICATIONS

SAR properties: roughness

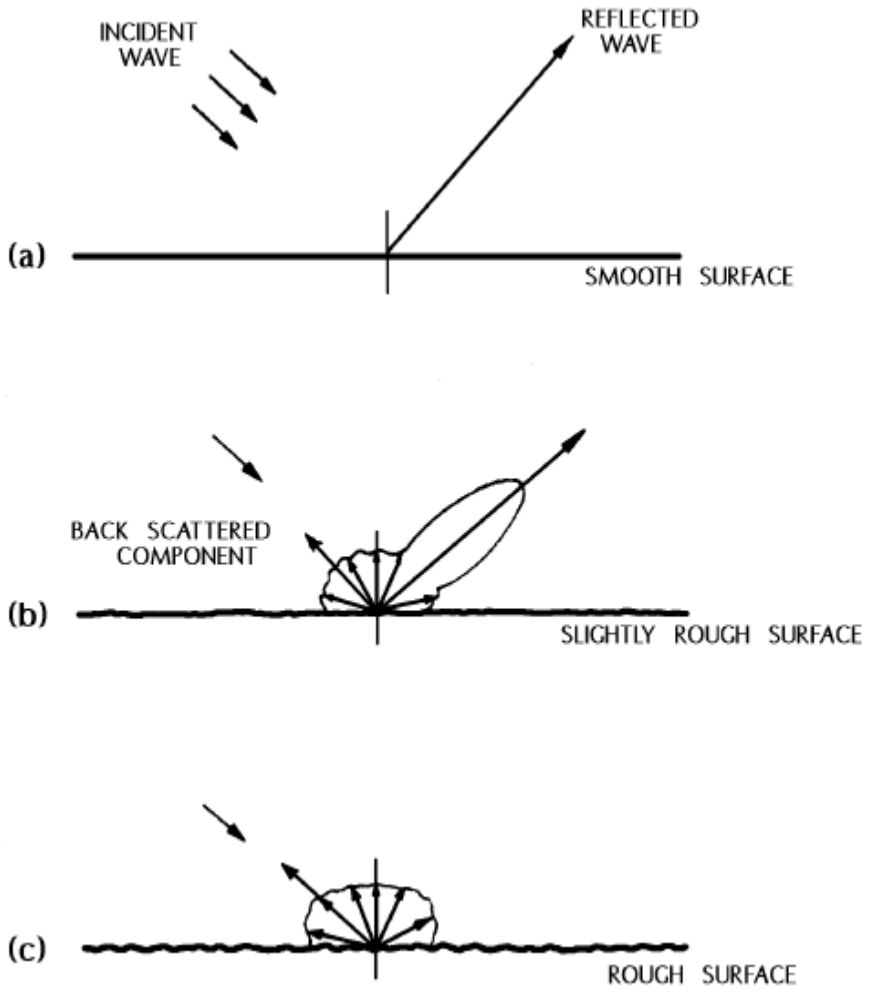
Roughness measuring device



A medicane (a rare Mediterranean cyclone), Jolina, off the coast of Libya. Key to classifying the storm as a medicane were images and data from the Copernicus Sentinel-1 mission, among others. This radar image from 18 March 2026, for example, was used to help identify the formation of a nearly-closed ring of wind near the sea surface and the windless eye-like feature in the centre of Medicane Jolina.

How SAR sees roughness?

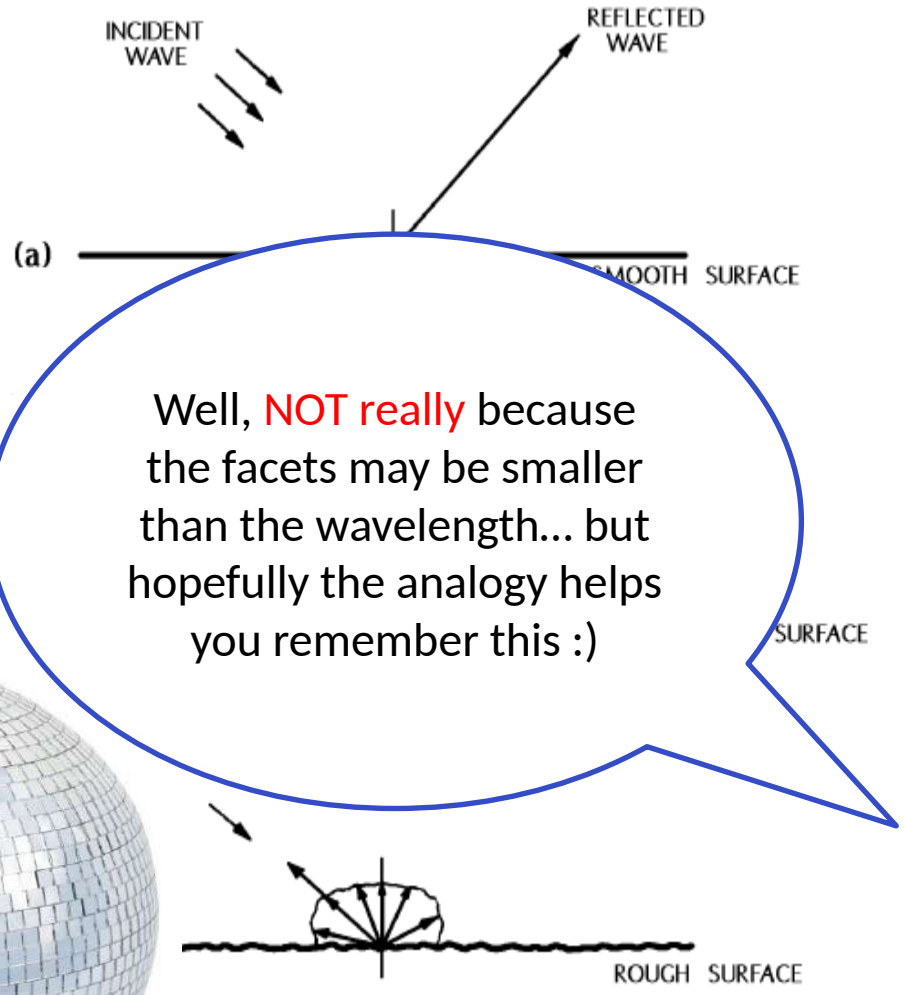
- ✓ To study the interaction of microwaves with water, we need a **scattering model**.
- ✓ Water is **impenetrable** by microwaves, therefore it is seen as a pure **surface**.
  - ✓ If the surface is **smooth**, the radiation will be reflected in the specular direction
  - ✓ If there is some surface **roughness** we can expect some return.
- ✓ In SAR images, we expect floods to be dark.



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    retu
- ✓ In
  be



That is like a discoball?

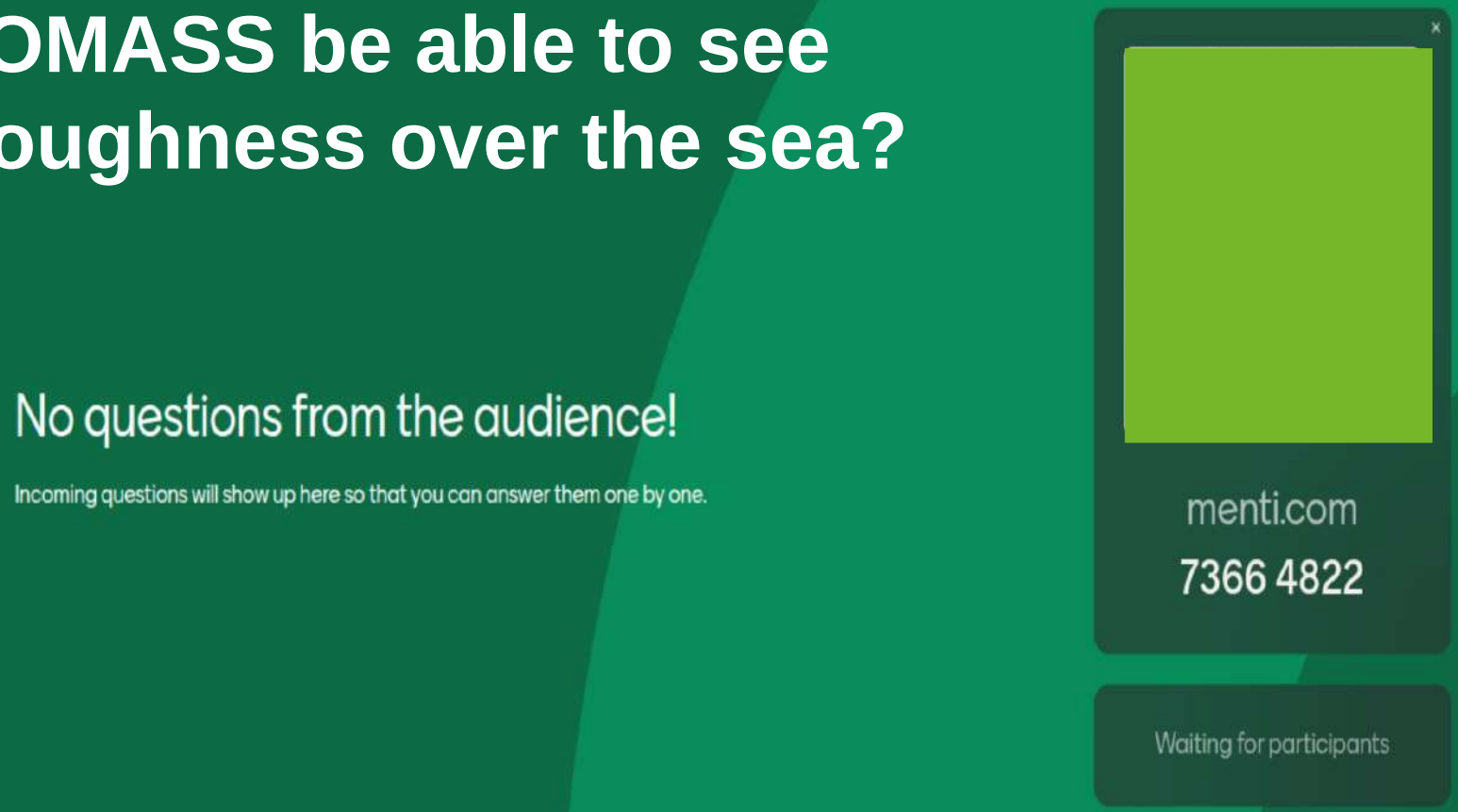




BE THE DIFFERENCE

What do you think?

Will BIOMASS be able to see wave roughness over the sea?



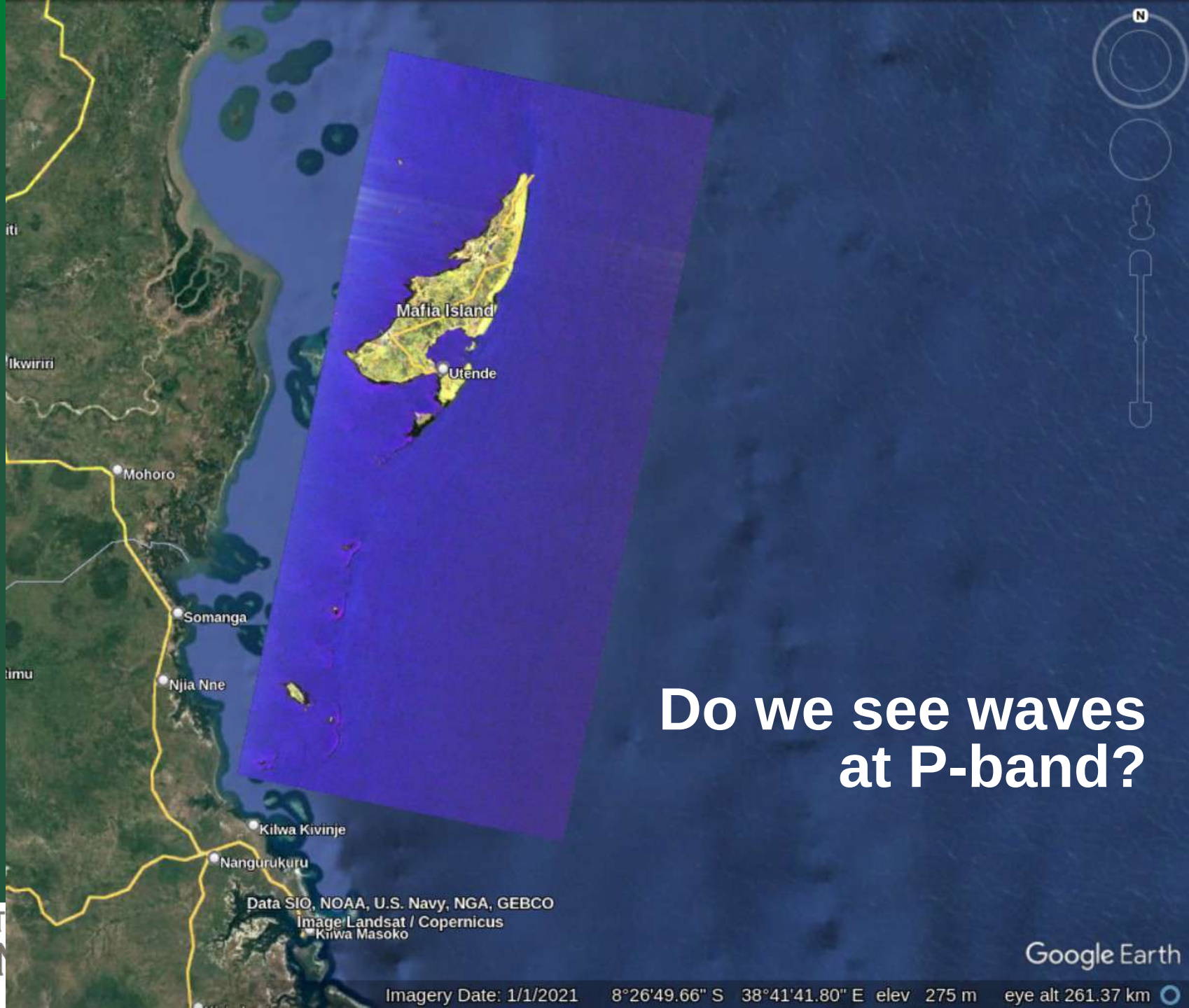
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BE THE DIFFERENCE



Do we see waves at P-band?

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Do we see waves at P-band?

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Moral of the story...

- ✓ Difference in roughness can lead to retrieval of parameters: waves.
- ✓ Roughness sometime is an annoyance because it produce variation in images that are not related on property that we may want to estimate…

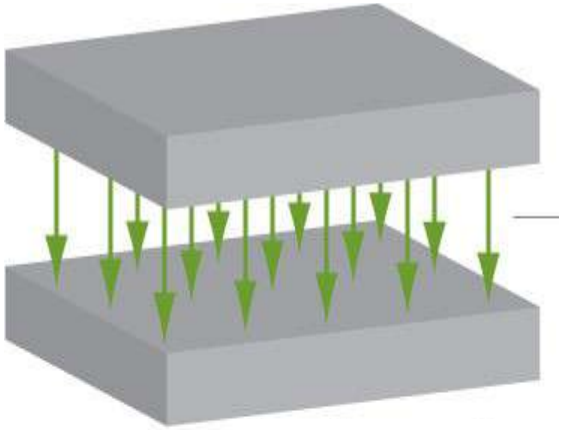
BE CAREFUL WHEN YOU ANALYSE SAR DATA AND DO NOT FORGET THAT ROUGHNESS MAY BE MODULATING YOUR PARAMETERS

SAR properties: dielectrics

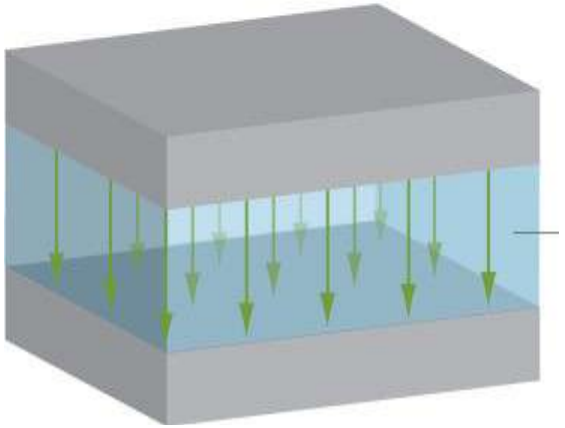
Dielectric constant explained

- It is the property of an object to **concentrate/absorb electric field**
- In principle, it is independent of the dimension
- It impact the capability of a body to emit at microwaves.
- But why do we care?!?!?

Low dielectric c.



High dielectric c.



Let’s think about a familiar example...

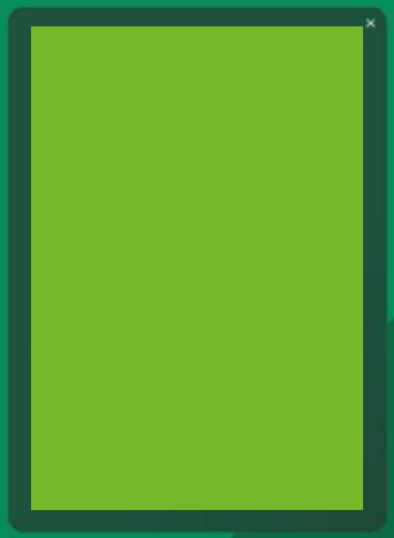


MICROWAVE FOOD FOR THOUGHT

Why microwaves oven are a nice example to explain why SAR is useful when it comes to measuring the dielectric constant?

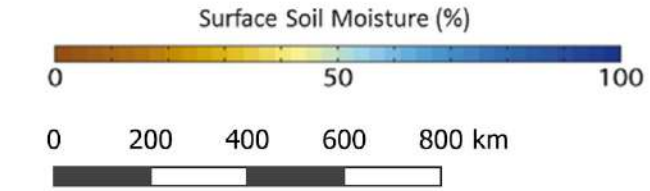
No questions from the audience!

Incoming questions will show up here so that you can answer them one by one.



Waiting for participants

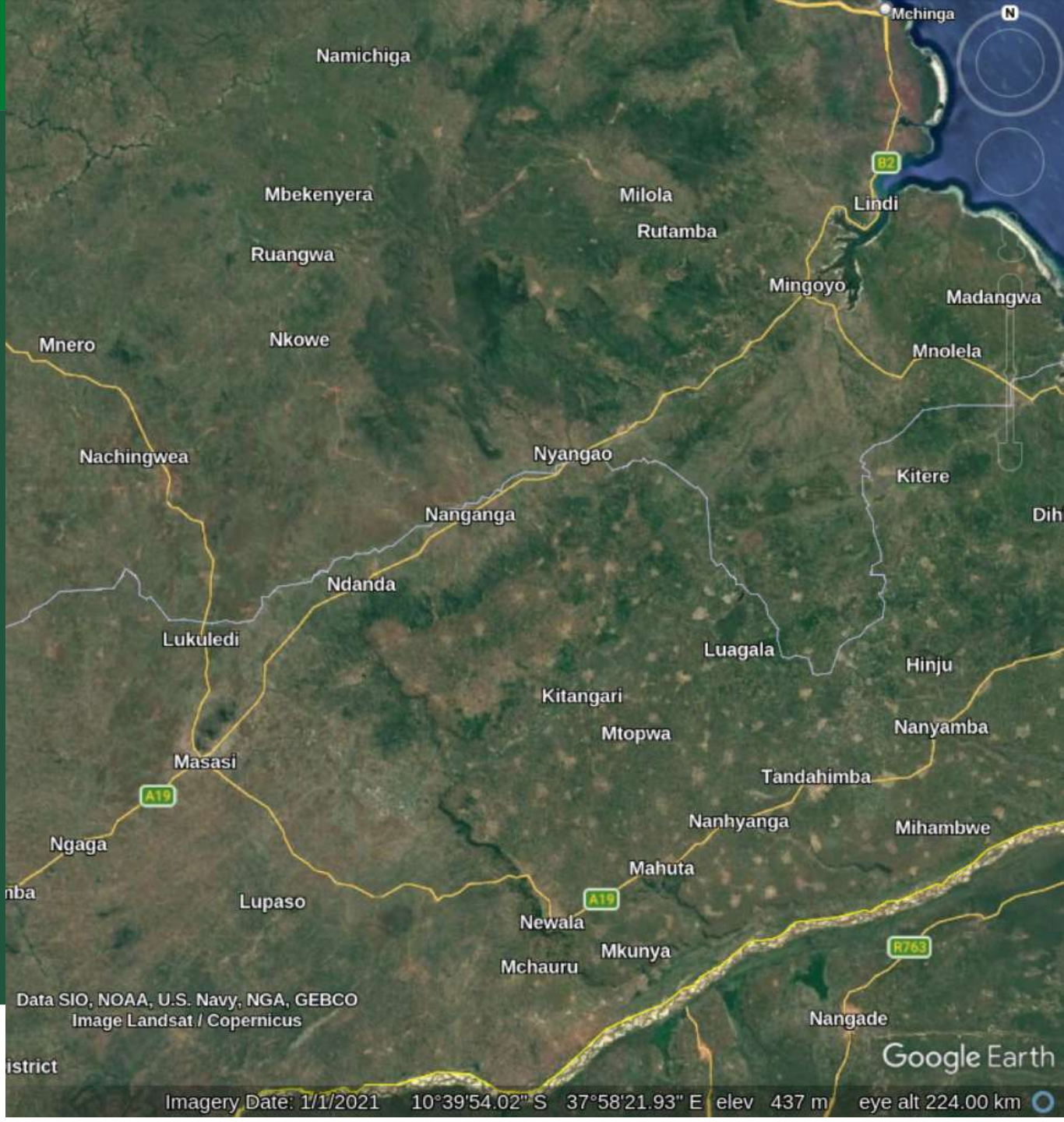
Soil moisture retrieval



Copernicus Global Land service, Daily Surface Soil Moisture based on Sentinel 1 24 Aug 2018

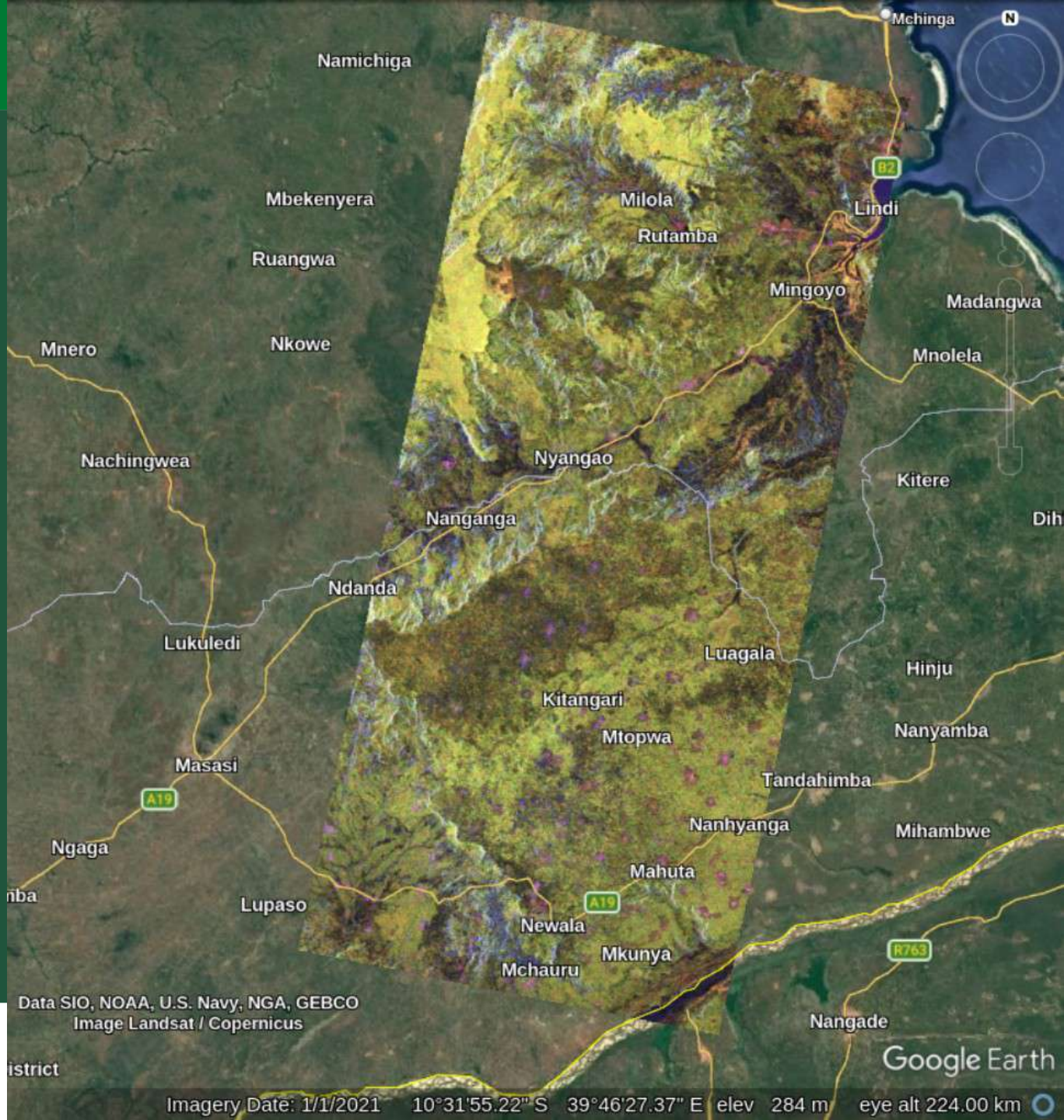
https://www.copernicus.eu/en/news/news/observer-monitoring-soil-moisture-space-copernicus

Are we sensitive to soil moisture at P-band? Tanzania.



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Are we sensitive to soil moisture at P-band? Tanzania.



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Moral of the story...

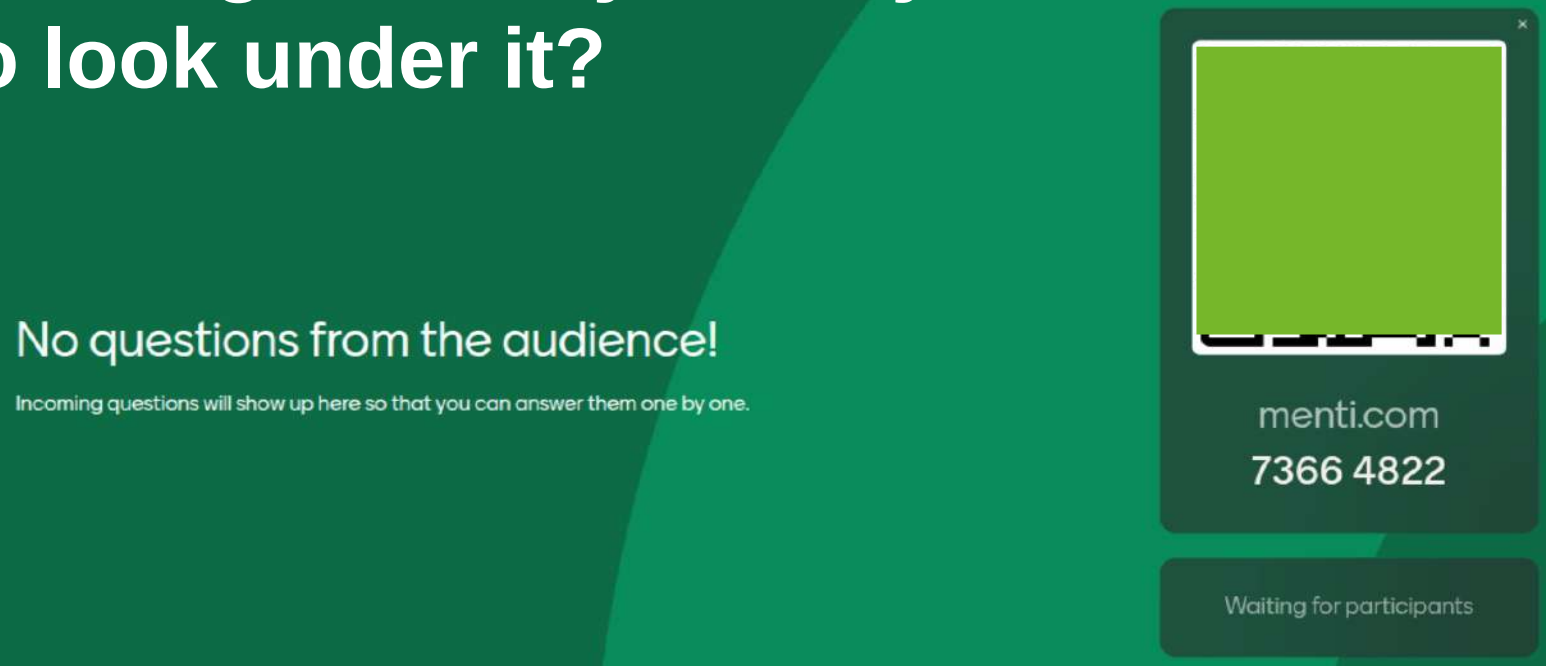
- ✓ Moisture is very valuable, but always keep in mind it may not be the only variable.
- ✓ Plants also have their dielectric constant and water content may change it

RADAR (AND BIOMASS) ARE VERY SENSITIVE TO WATER CONTENT OF THINGS… RETRIEVE IT, OR REMOVE IT

SAR properties: penetration

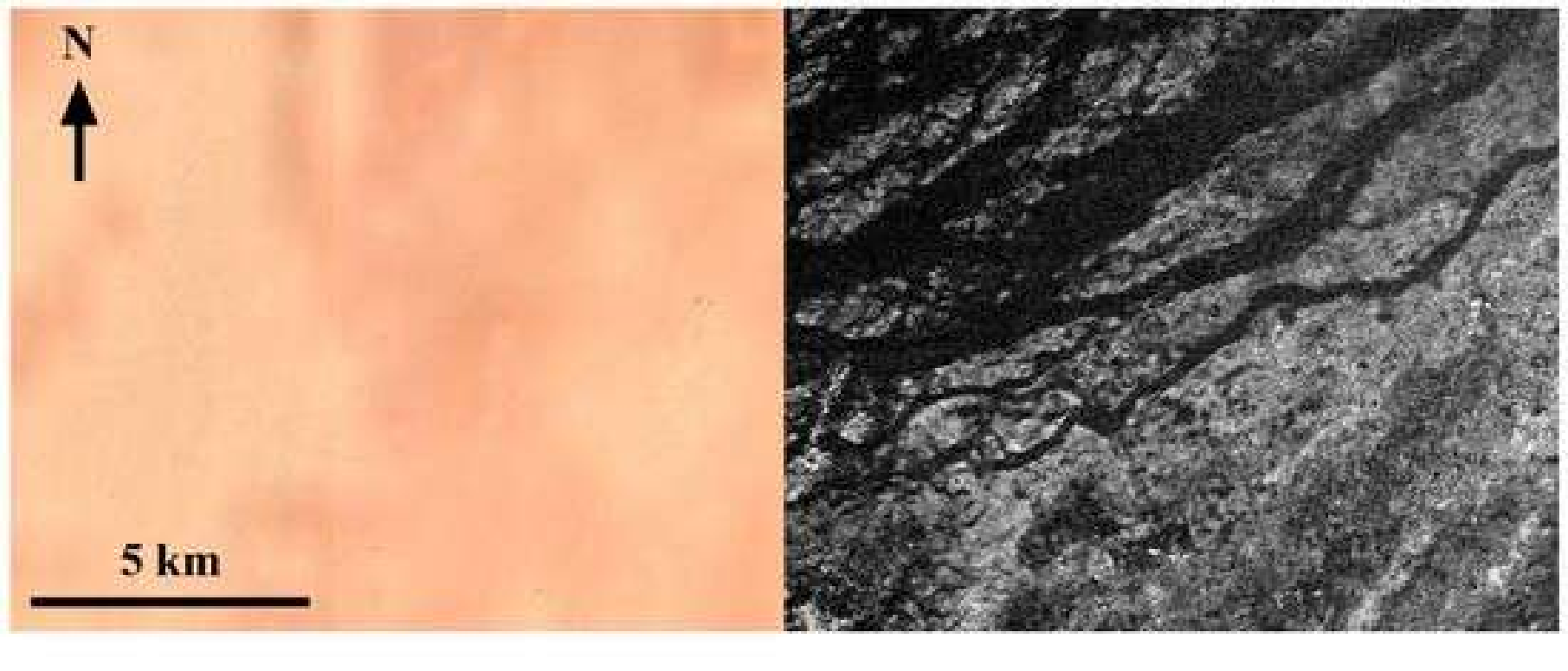
Interesting targets…

What are targets that you may want to look under it?



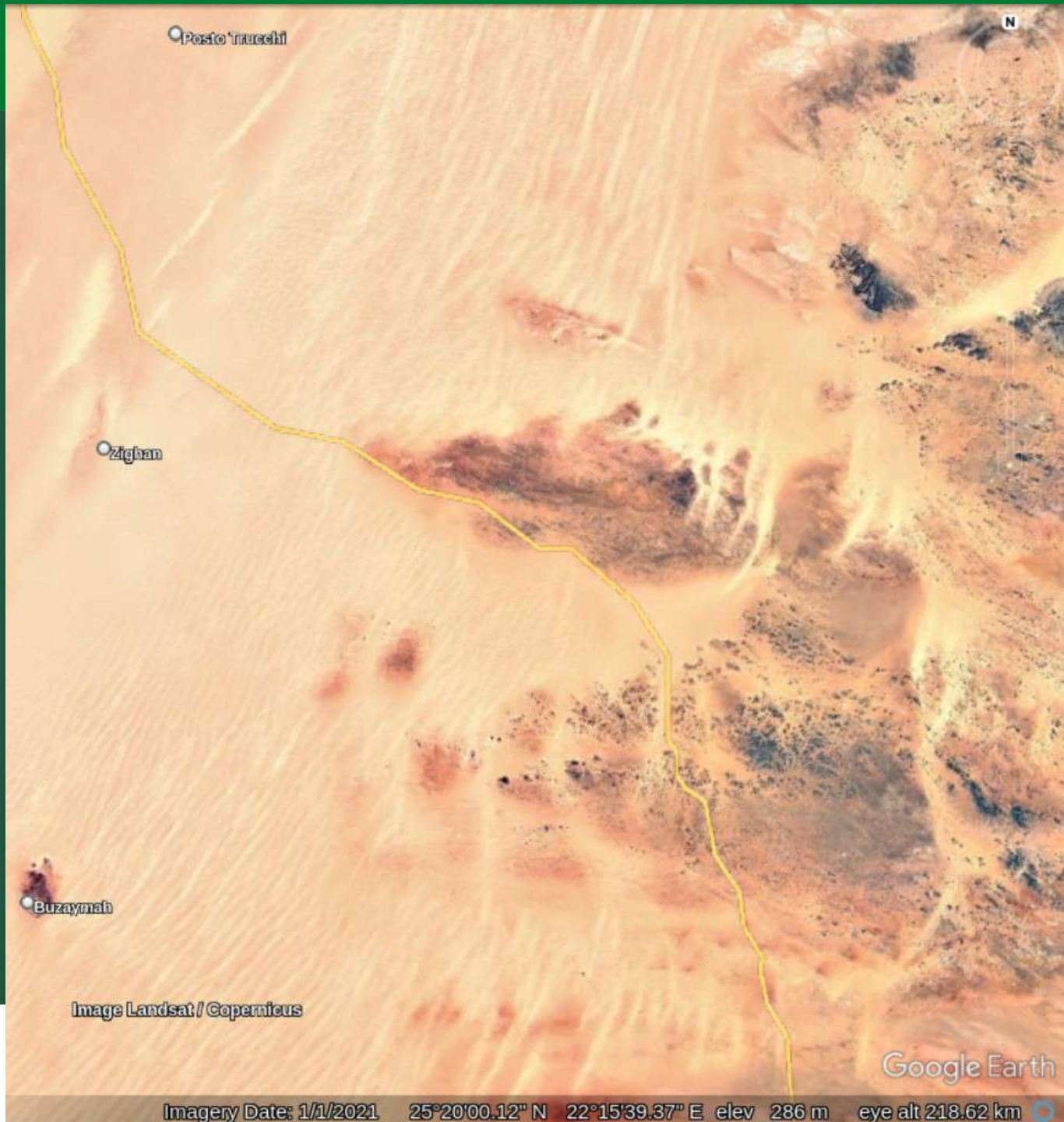
Penetration in media: desert

(Left) SPOT (Satellite Pour l'Observation de la Terre) image of part of the Selima Sand Sheet located in southern Egypt (22.55° N–29.28° E).
(Right) Radar image from Japanese polarimetric L-band synthetic aperture radar (PALSAR) of the same area, revealing numerous palaeochannels hidden by a meter-thick layer of aeolian sand.



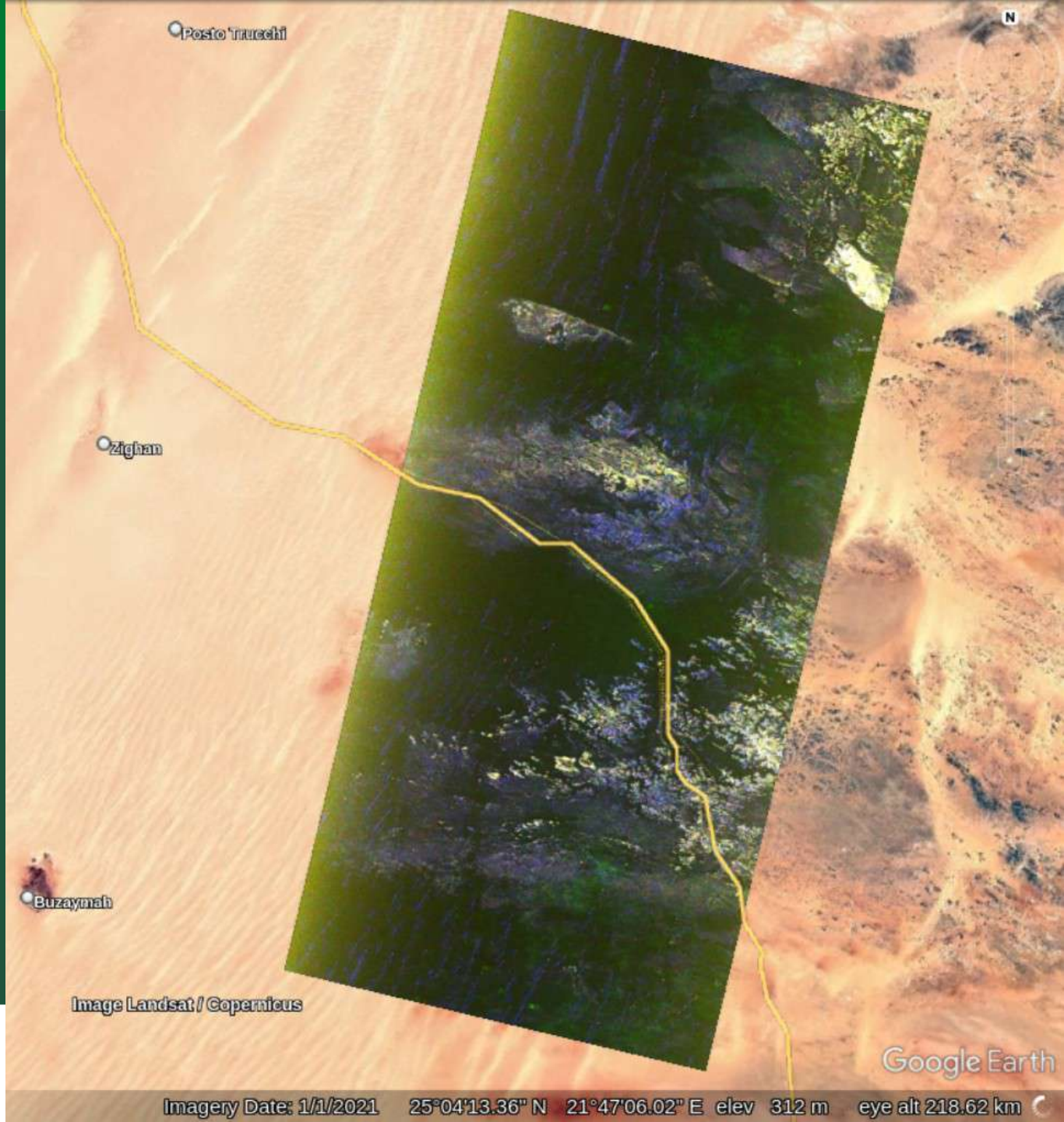
Paillou, P. Mapping Palaeohydrography in Deserts: Contribution from Space-Borne Imaging Radar. Water 2017, 9, 194. https://doi.org/10.3390/w9030194

Who wants a P-band Dessert? Egypt.



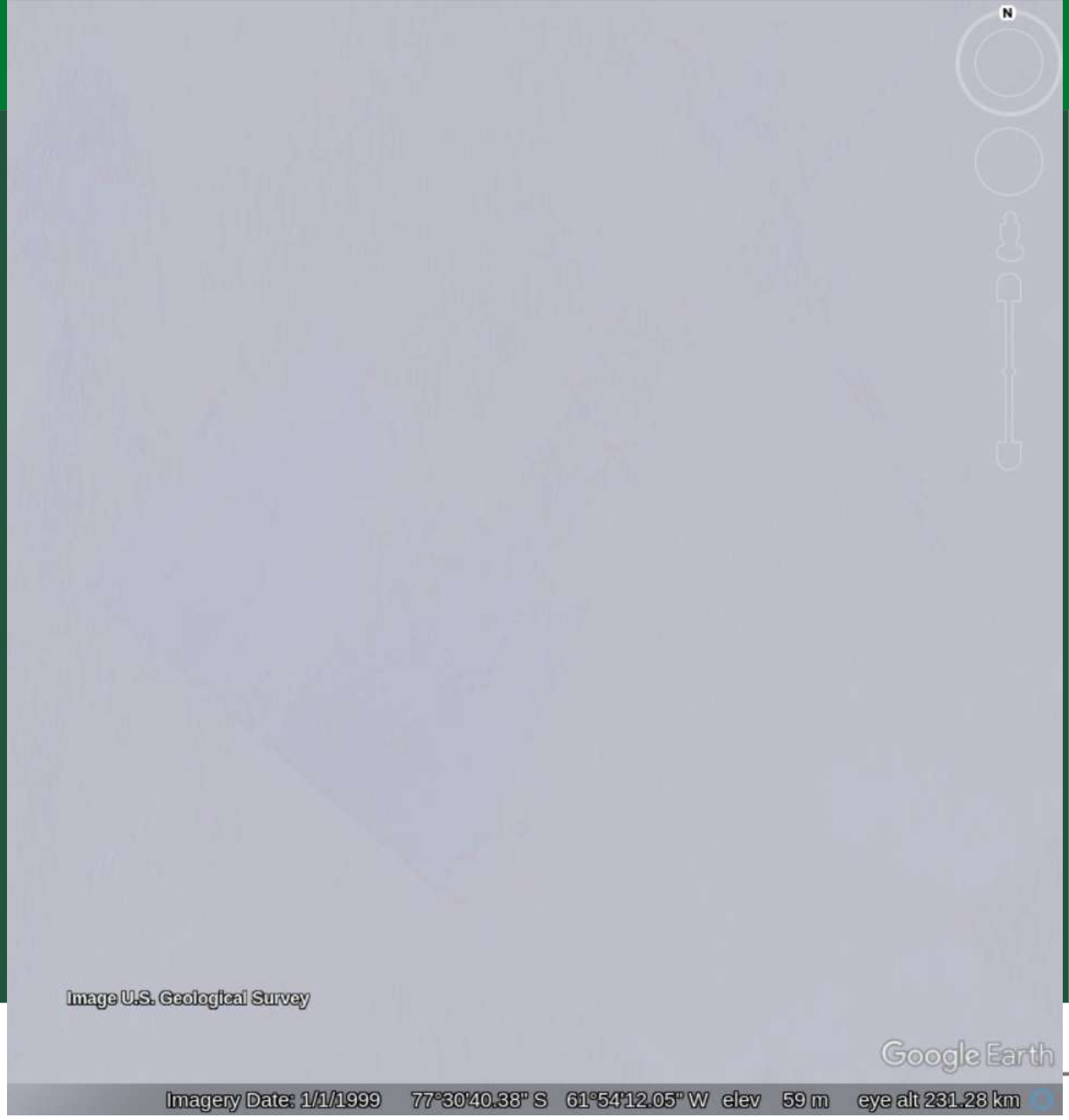
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Who wants a P-band Dessert? Egypt.



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More on the theme of desserts… Antarctica.



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More on the theme of desserts… Antarctica.



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Moral of the story...

- ✓ We can use SAR to look at things that are hidden from passive optical systems.

TAKE ADVANTAGE OF BIOMASS EXTENDED PENETRATION CAPABILTIES… IT IS A UNIQUE FEATURE

PolSAR

Playing with antenna orientations

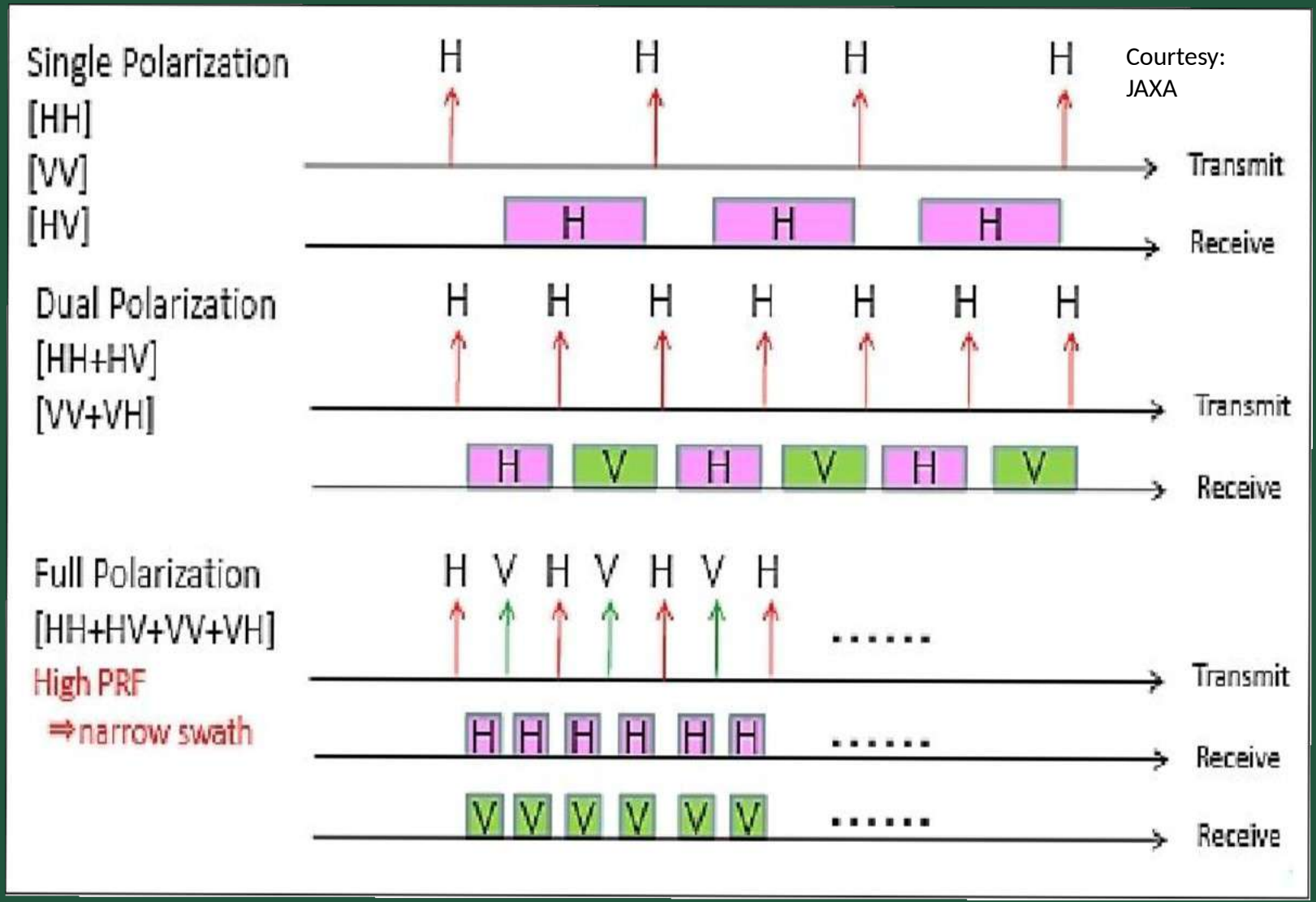


Ubiquitous polarimetry

Do you know of any example of use of polarimetry? Outside the radar context too.



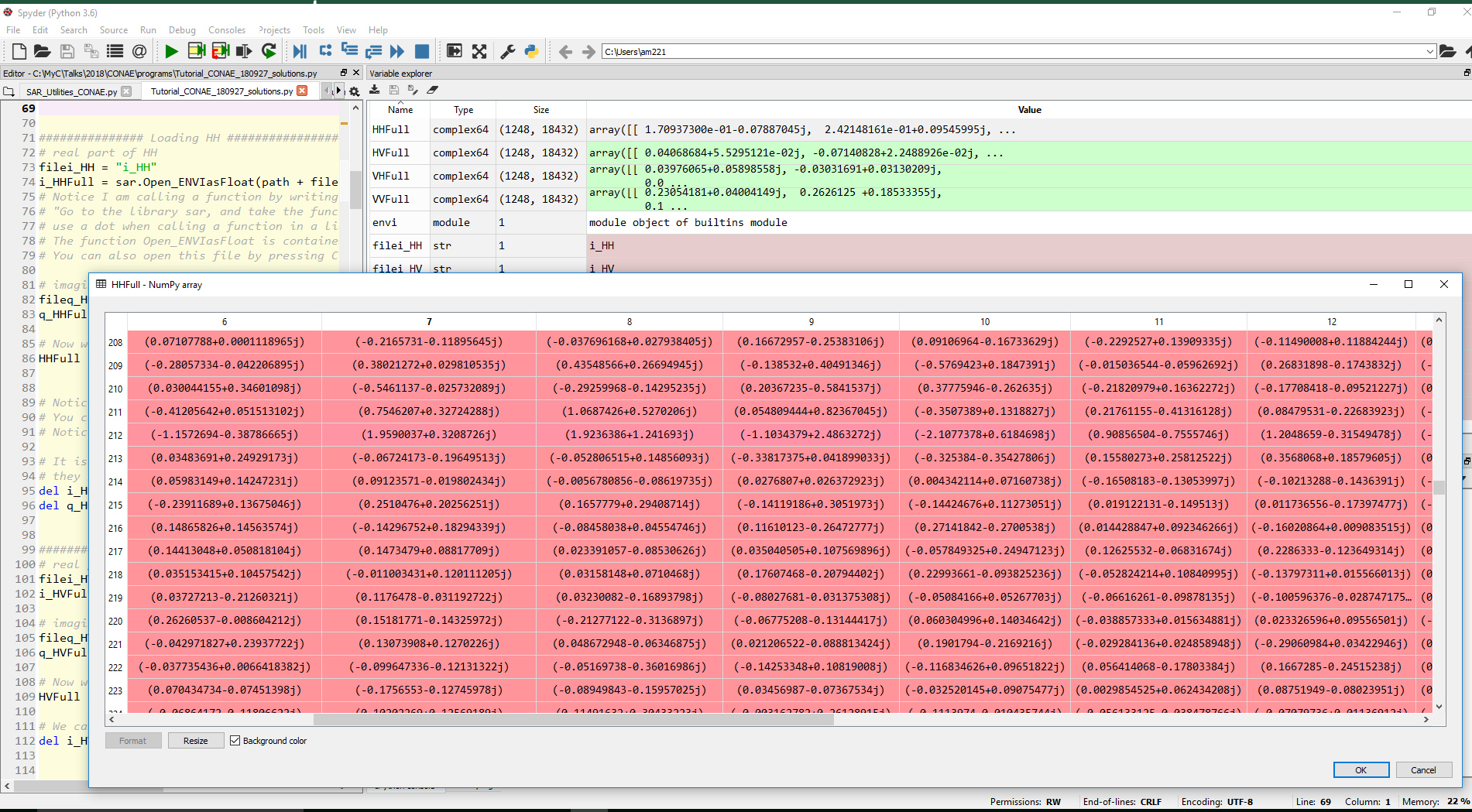
Playing with antenna orientations



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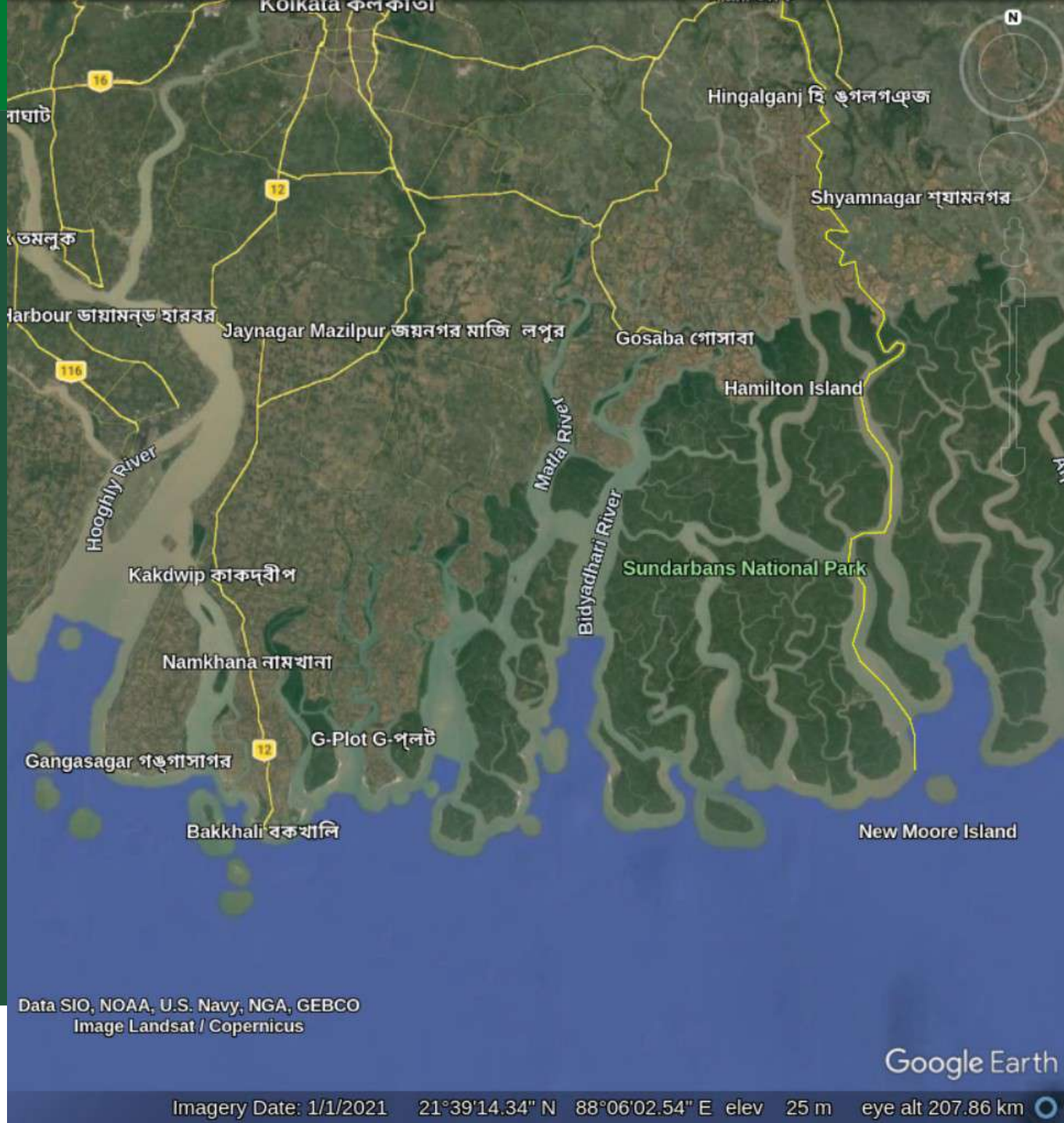
Complex… but not impossible

Data are stored in complex form, that is real plus imaginary



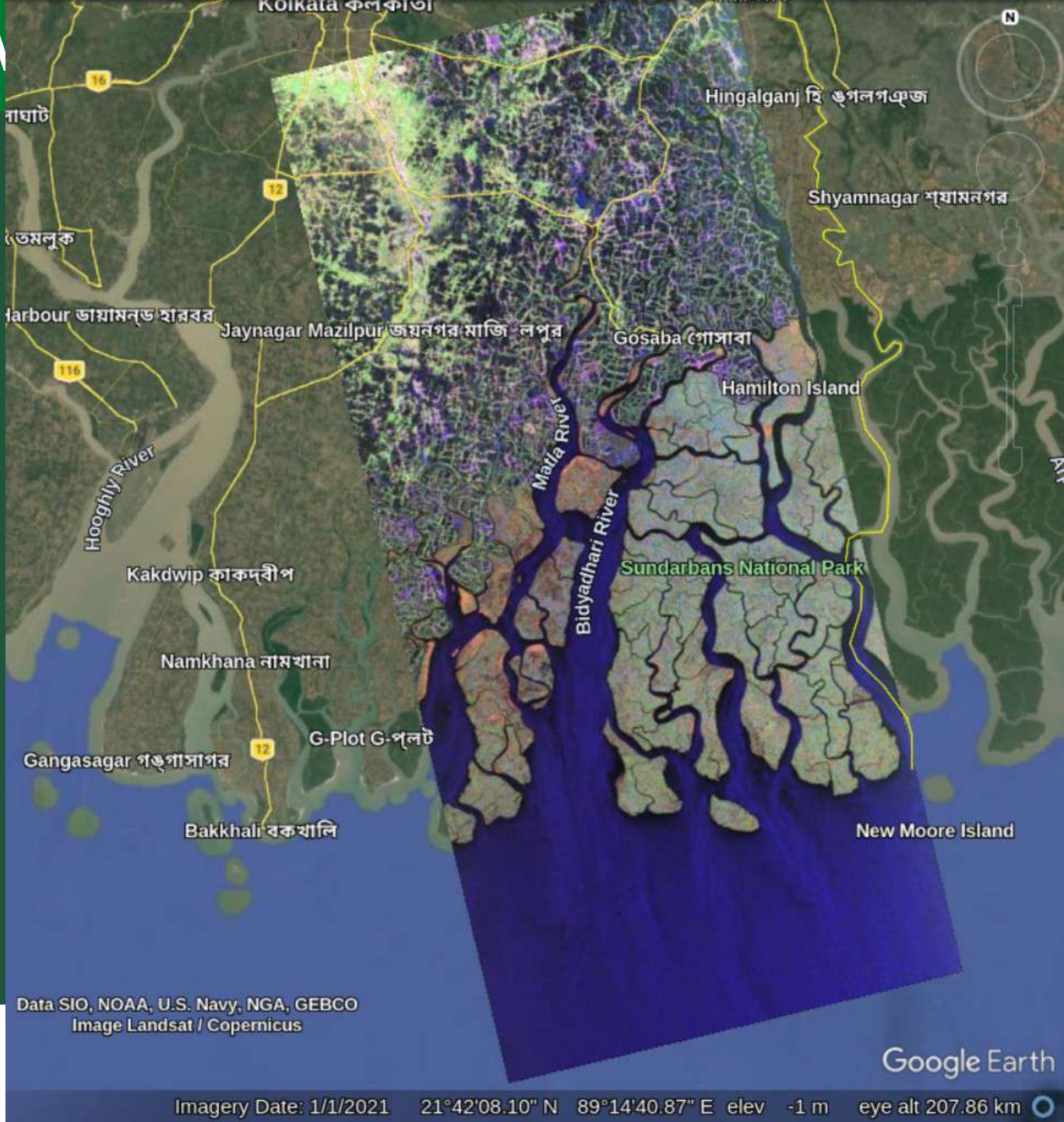
	6	7	8	9	10	11	12
208	(0.07107788+0.00011189 65j)	(-0.2165731-0.11895645j)	(-0.037696168+0.027938405j)	(0.16672957-0.25383106j)	(0.09106964-0.16733629j)	(-0.2292527+0.13909335j)	(-0.11490008+0.11884244j)
209	(-0.28057334-0.042206895j)	(0.38021272+0.029810535j)	(0.43548566+0.26694945j)	(-0.138532+0.40491346j)	(-0.5769423+0.1847391j)	(-0.015036544-0.05962692j)	(0.26831898-0.1743832j)
210	(0.030044155+0.34601098j)	(-0.5461137-0.025732089j)	(-0.29259968-0.14295235j)	(0.20367235-0.5841537j)	(0.37775946-0.262635j)	(-0.21820979+0.16362272j)	(-0.17708418-0.09521227j)
211	(-0.41205642+0.051513102j)	(0.7546207+0.32724288j)	(1.0687426+0.5270206j)	(0.054809444+0.82367045j)	(-0.3507389+0.1318827j)	(0.21761155-0.41316128j)	(0.08479531-0.22683923j)
212	(-1.1572694-0.38786665j)	(1.9590037+0.3208726j)	(1.9236386+1.241693j)	(-1.1034379+2.4863272j)	(-2.1077378+0.6184698j)	(0.90856504-0.7555746j)	(1.2048659-0.31549478j)
213	(0.03483691+0.24929173j)	(-0.06724173-0.19649513j)	(-0.052806515+0.14856093j)	(-0.33817375+0.041899033j)	(-0.325384-0.35427806j)	(0.15580273+0.25812522j)	(0.3568068+0.18579605j)
214	(0.05983149+0.14247231j)	(0.09123571-0.019802434j)	(-0.0056780856-0.08619735j)	(0.0276807+0.026372923j)	(0.004342114+0.07160738j)	(-0.16508183-0.13053997j)	(-0.10213288-0.1436391j)
215	(-0.23911689+0.13675046j)	(0.2510476+0.20256251j)	(0.1657779+0.29408714j)	(-0.14119186+0.3051973j)	(-0.14424676+0.11273051j)	(0.019122131-0.149513j)	(0.011736556-0.17397477j)
216	(0.14865826+0.14563574j)	(-0.14296752+0.18294339j)	(-0.08458038+0.04554746j)	(0.11610123-0.26472777j)	(0.27141842-0.2700538j)	(0.014428847+0.092346266j)	(-0.16020864+0.009083515j)
217	(0.14413048+0.050818104j)	(0.1473479+0.08817709j)	(0.023391057-0.08530626j)	(0.035040505+0.107569896j)	(-0.057849325+0.24947123j)	(0.12625532-0.06831674j)	(0.2286333-0.123649314j)
218	(0.035153415+0.10457542j)	(-0.011003431+0.120111205j)	(0.03158148+0.0710468j)	(0.17607468-0.20794402j)	(0.22993661-0.093825236j)	(-0.052824214+0.10840995j)	(-0.13797311+0.015566013j)
219	(0.03727213-0.21260321j)	(0.1176478-0.031192722j)	(0.03230082-0.16893798j)	(-0.08027681-0.031375308j)	(-0.05084166+0.05267703j)	(-0.06616261-0.09878135j)	(-0.100596376-0.028747175…
220	(0.26260537-0.008604212j)	(0.15181771-0.14325972j)	(-0.21277122-0.3136897j)	(-0.06775208-0.13144417j)	(0.060304996+0.14034642j)	(-0.038857333+0.015634881j)	(0.023326596+0.09556501j)
221	(-0.042971827+0.23937722j)	(0.13073908+0.1270226j)	(0.048672948-0.06346875j)	(0.021206522-0.088813424j)	(0.1901794-0.2169216j)	(-0.029284136+0.024858948j)	(-0.29060984+0.03422946j)
222	(-0.037735436+0.0066418382j)	(-0.099647336-0.12131322j)	(-0.05169738-0.36016986j)	(-0.14253348+0.10819008j)	(-0.116834626+0.09651822j)	(0.056414068-0.17803384j)	(0.1667285-0.24515238j)
223	(0.070434734-0.07451398j)	(-0.1756553-0.12745978j)	(-0.08949843-0.15957025j)	(0.03456987-0.07367534j)	(-0.032520145+0.09075477j)	(0.0029854525+0.062434208j)	(0.08751949-0.08023951j)

Mangroves at P-band. Bangladesh



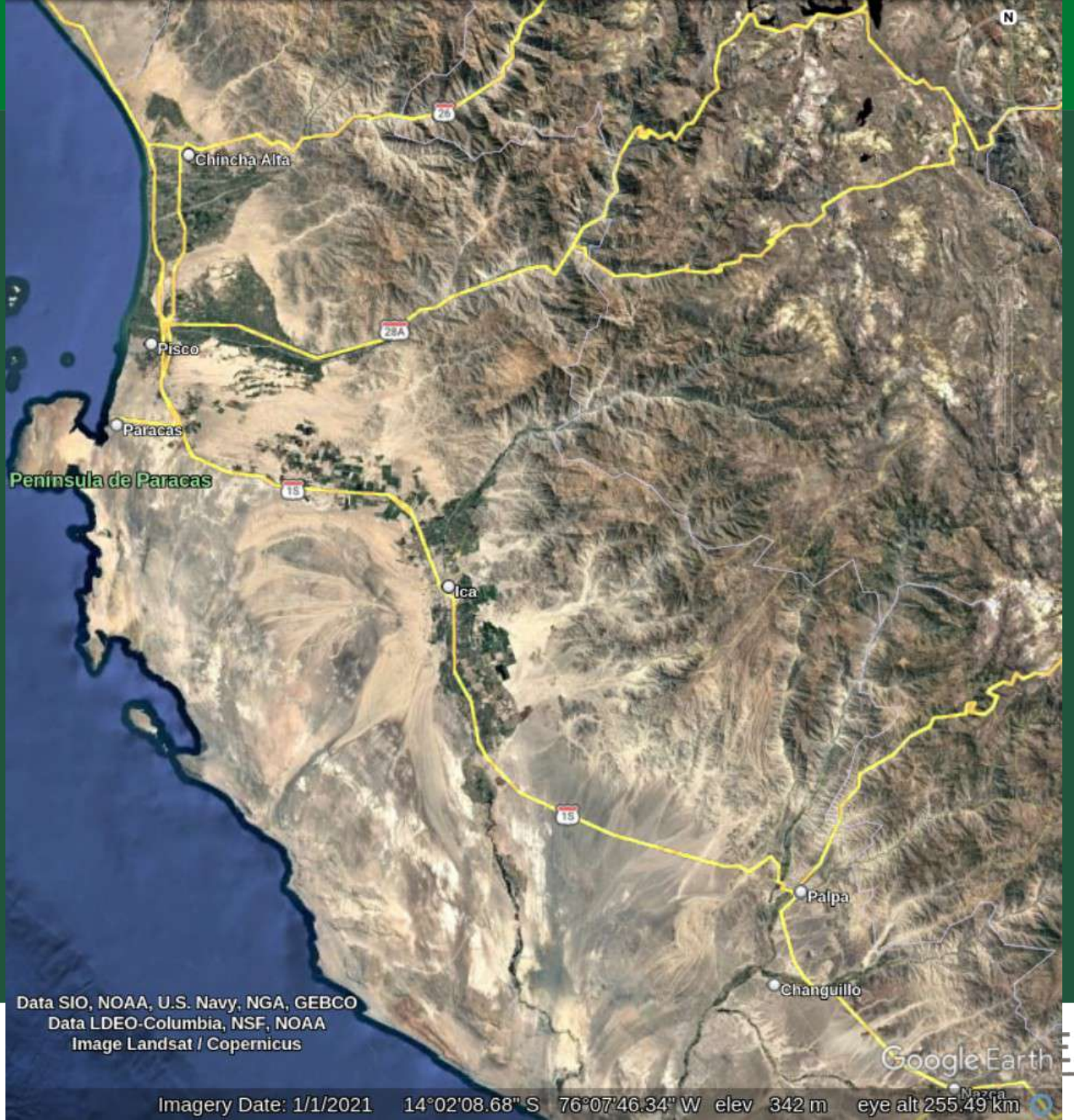
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Mangroves at P-band. Bangladesh



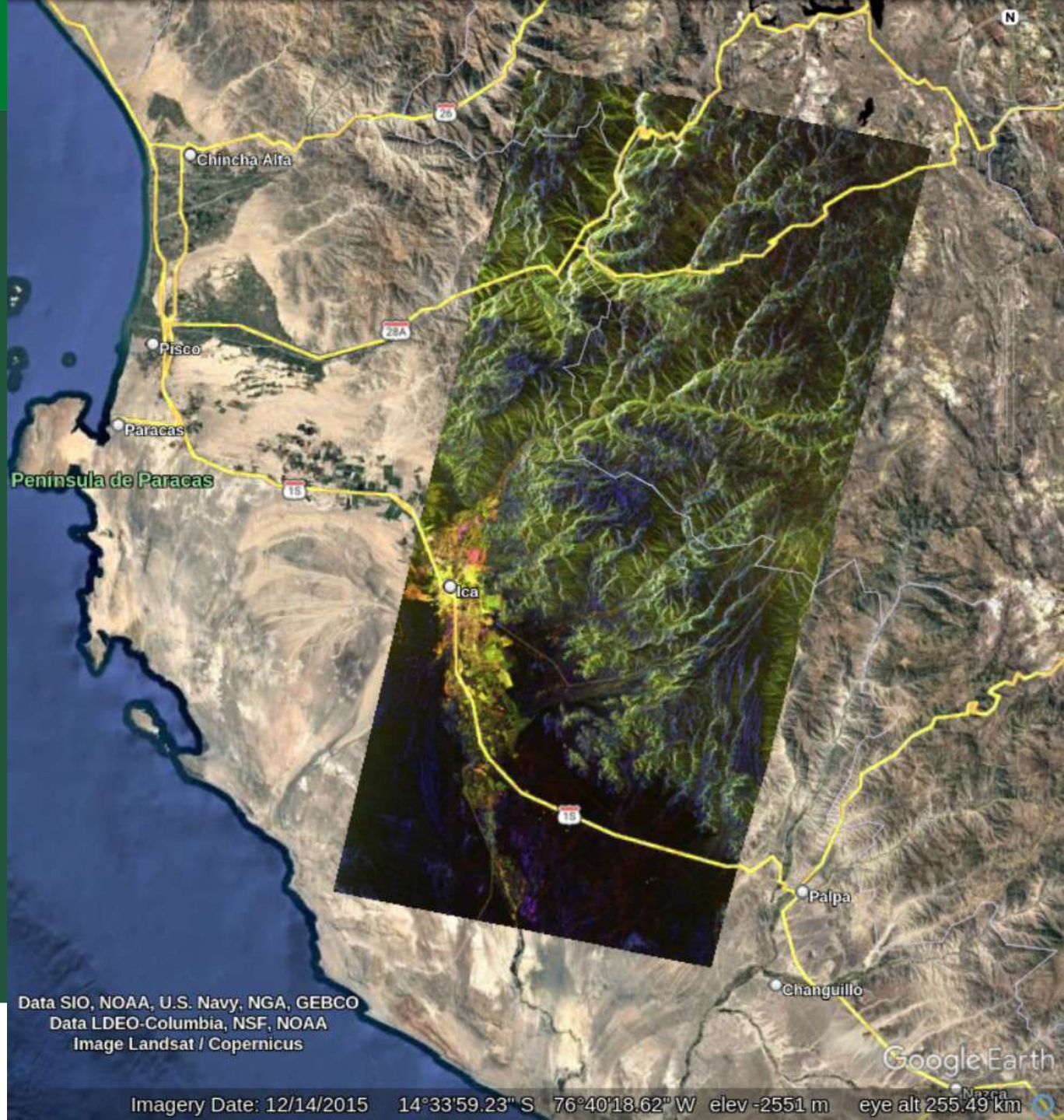
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The eye in the sky?!?! Nazca.



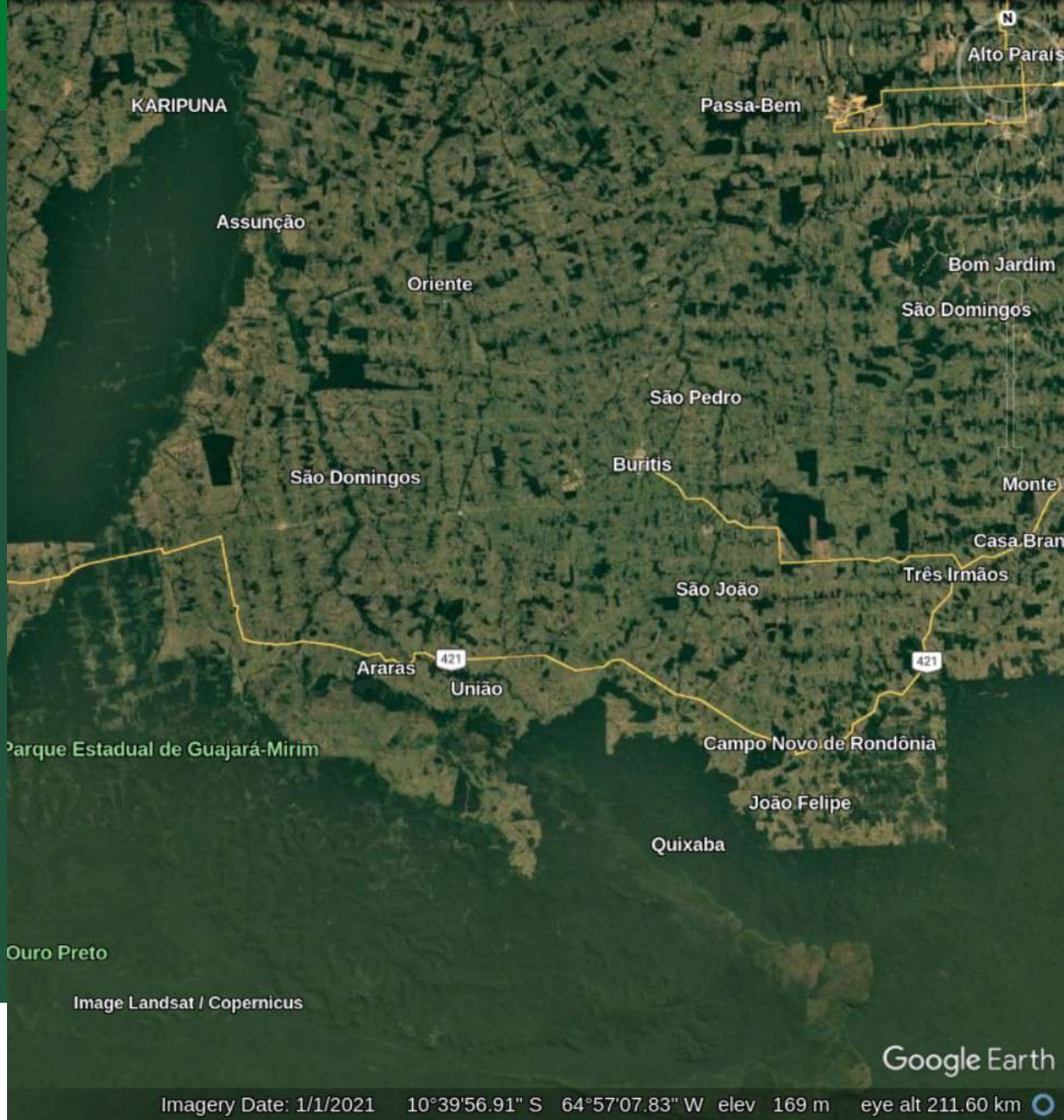
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The eye in the sky?!?! Nazca.



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Now you see me… now you don’t… Pampas.



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Now you see me… now you don't… Pampas.



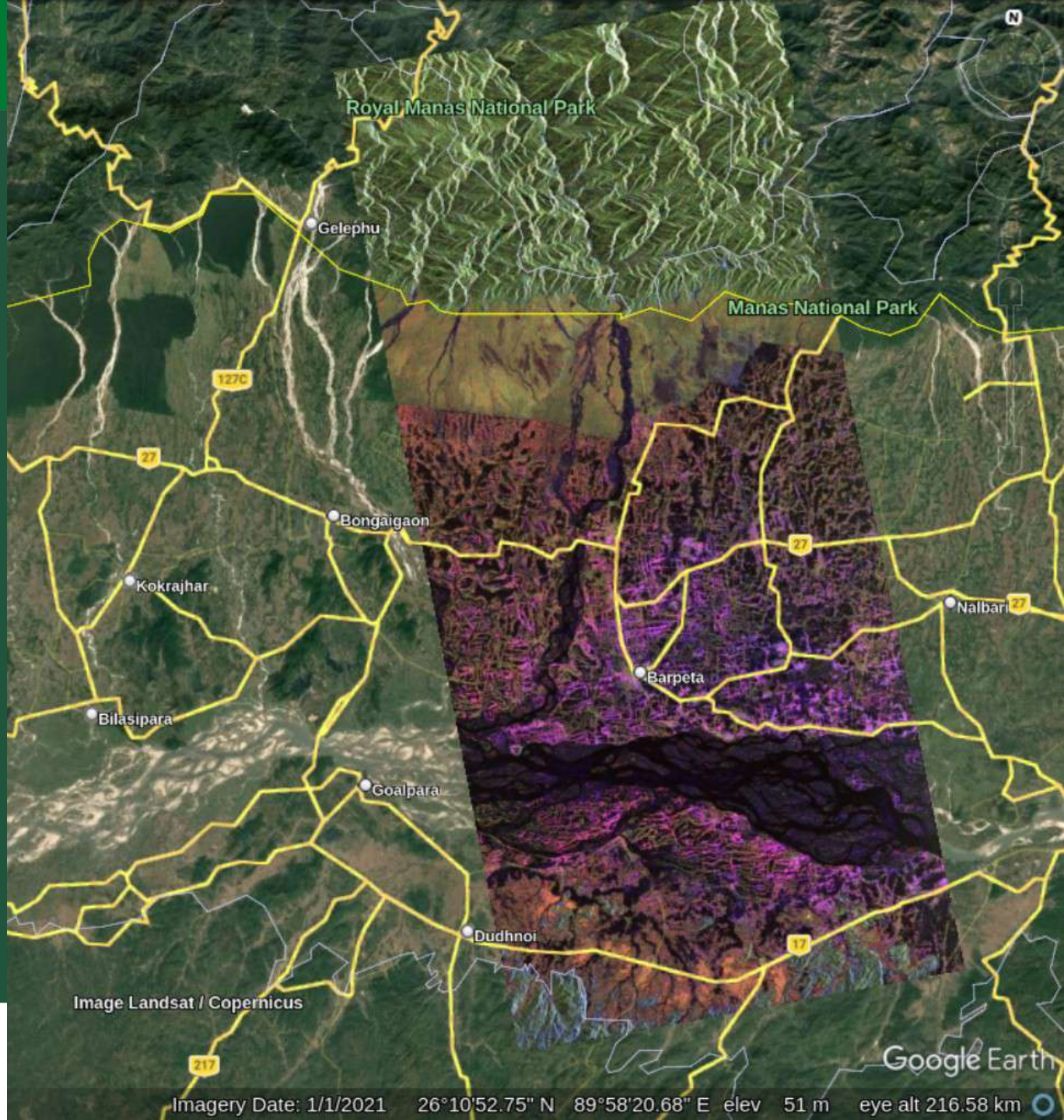
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Famous rivers... Gangi.



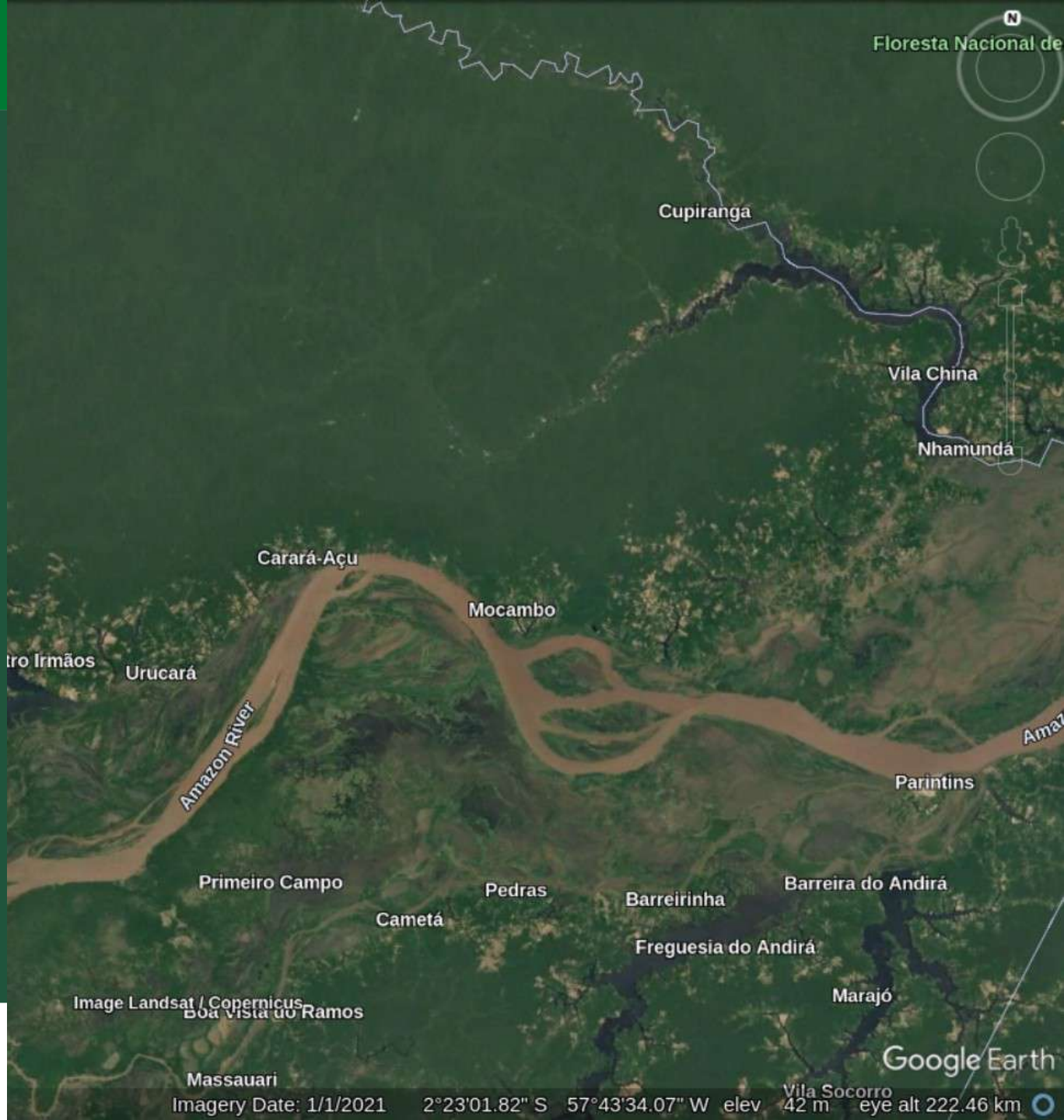
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Famous rivers... Gangi.



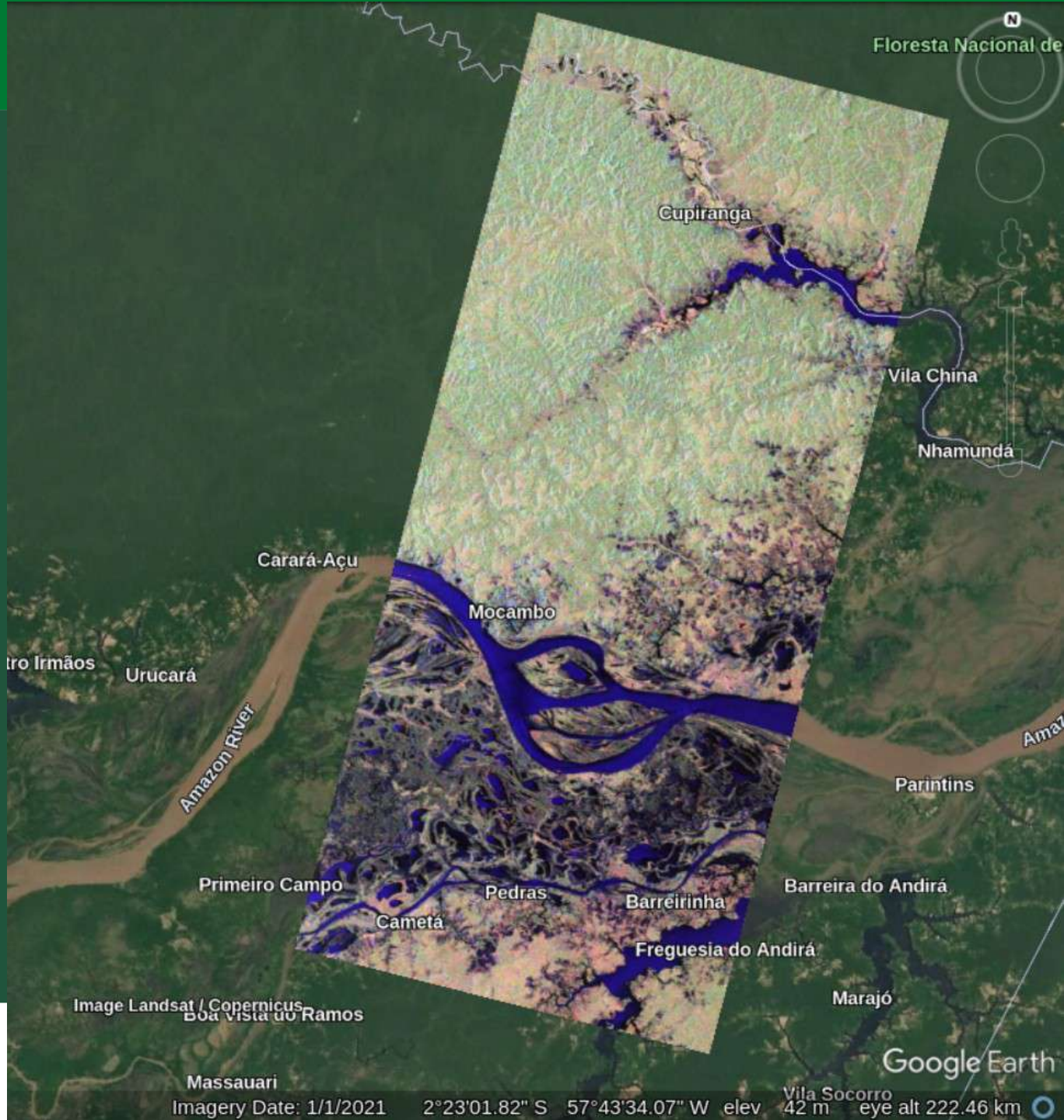
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Another famous river... Amazon.



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Another famous river... Amazon.



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Moral of the story...

- ✓ PolSAR allow us to have a physical understanding of the target
- ✓ It allow to improve our detection/classification/retrivial

TRY TO EXPLOIT THE POLARIOMETRIC INFORMATION CAREFULLY… IF YOU ARE UNSURE LOOK AT SOME ESA TUTORIAL ONLINE

PolInSAR

Combining PolSAR and InSAR in a forest



Better than the sum of the two…

Why combining PolSAR and InSAR in a forest may be useful? What do we expect to see?



Example of PolInSAR

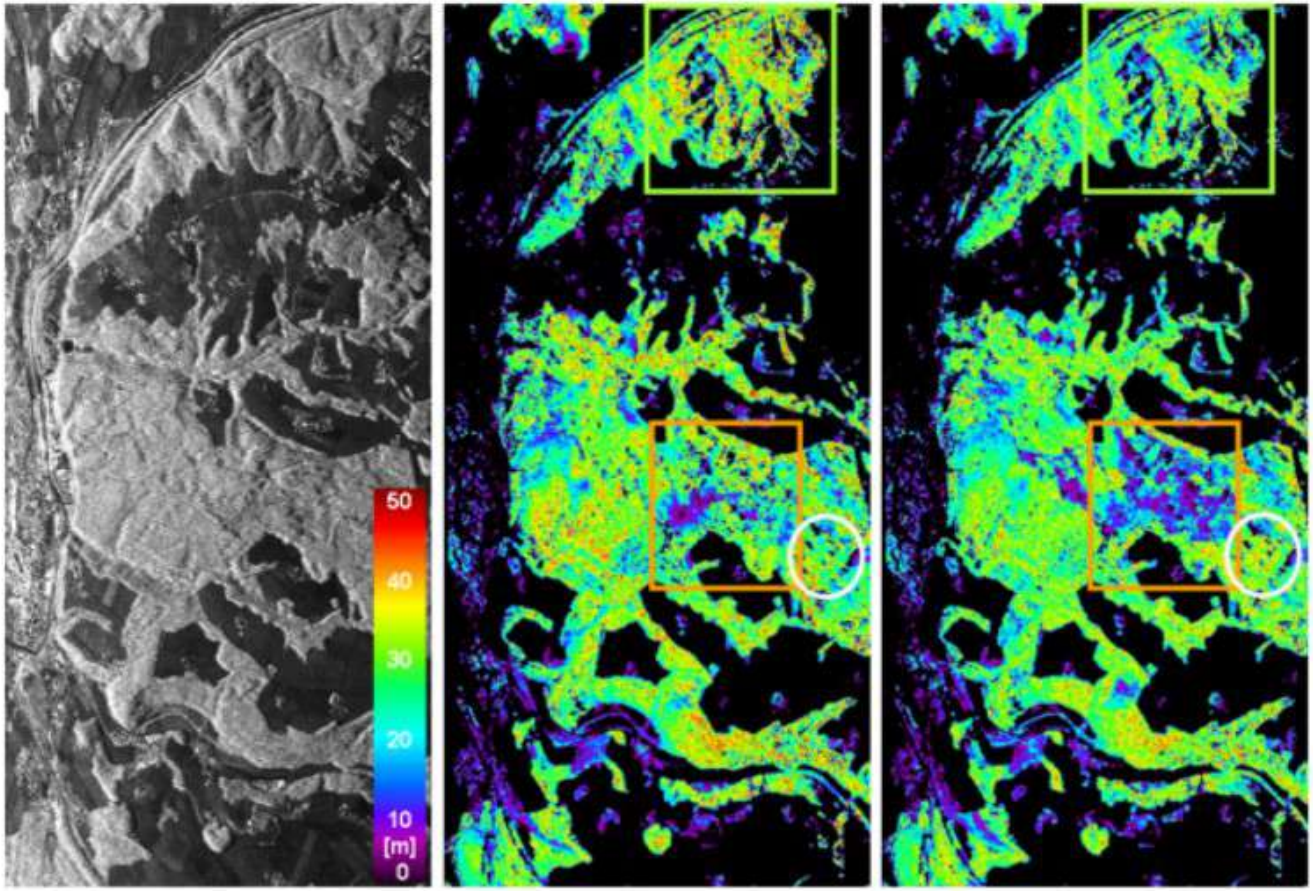
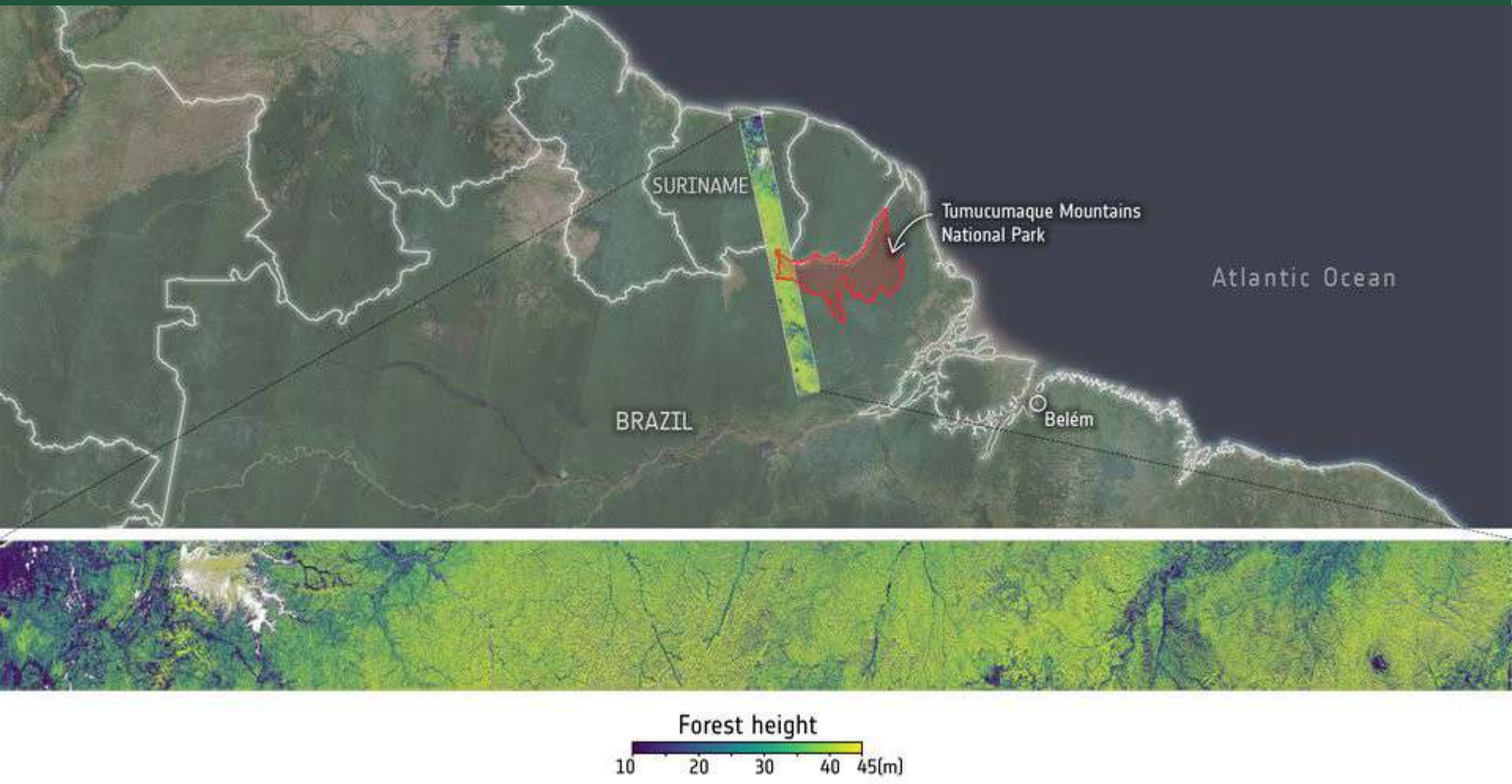


Fig. 1. L-band HV intensity image of the Traunstein test site (left). Forest height map computed from Pol-InSAR data in 2003 (middle) and 2008 (right).

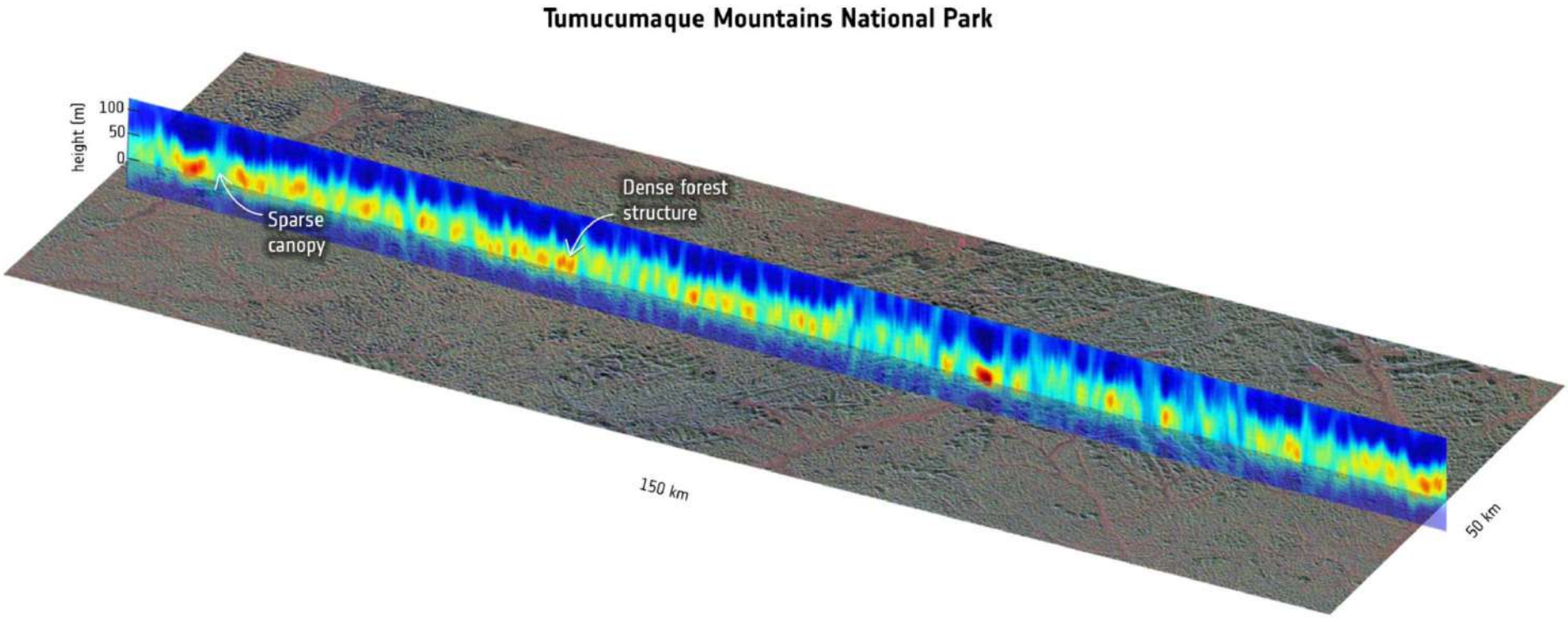
TomoSAR

BIOMASS TomoSAR



ESA - Brazil gears up to harness ESA's https://www.esa.int/Applications/Observing_the_Earth/FutureEO/Biomass/Brazil_gears_up_to_harness_ESA_s_Biomass_dataBiomass data

BIOMASS TomoSAR



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Now what?

How would you like to use BIOMASS now?




Thank you for your attention!